



THE CO-OPERATIVE UNIVERSITY OF KENYA

SCHOOL OF COMPUTING AND MATHEMATICS

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY

SHES – SMART HEALTH EXPERT SYSTEM: AN AI-DRIVEN CLINICAL DECISION SUPPORT AND PREVENTIVE HEALTHCARE PLATFORM FOR KENYA

A Project Report Submitted to the School of Computing and Mathematics in Partial
Fulfilment of the Requirements for the Award of the Degree of Bachelor of Science in
Software Engineering

BY

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DECLARATION

Student Declaration

I, Stephen Ombuya, hereby declare that this project report titled "SHES – Smart Health Expert System: An AI-Driven Clinical Decision Support and Preventive Healthcare Platform for Kenya" was the result of my own original research and work. It had not been presented for a degree or any other academic award in any other university or institution. All sources of information used in this report were duly acknowledged and referenced in accordance with the APA referencing style.

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Supervisor's Declaration

This project report was submitted for examination with my approval as the University supervisor.

Supervisor's Name: Dr. Duncan Nyale

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Date: _____

DEDICATION

To every Kenyan who has ever waited too long for medical help that should have been within reach —

This work was conceived for you.

To my family, whose unwavering support provided the foundation upon which this project stood. To my lecturers at The Co-operative University of Kenya, whose guidance sharpened both my technical and intellectual capacities. And to the open-source global health informatics community, whose published research and clinical guidelines informed the clinical logic at the core of this system.

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Special recognition was given to the developers and maintainers of the open-source frameworks and libraries that powered SHES, including the Django Project, the React development team, the PostgreSQL Global Development Group, and contributors to Python's Scikit-learn library. Their freely available tools made the realization of an enterprise-grade health platform achievable within the constraints of an undergraduate project.

Finally, I acknowledged the Kenya Ministry of Health for publicly disseminating the Kenya National Clinical Guidelines and the Kenya Essential Medicines List, which provided the authoritative clinical knowledge base upon which the SHES inference engine was grounded.

ABSTRACT

Kenya's healthcare sector continues to experience significant challenges, including a low doctor-to-patient ratio, inadequate access to quality healthcare in rural areas, rising cases of chronic illnesses, and congestion in public health facilities, all of which contribute to delayed diagnosis, poor medication adherence, and limited access to timely clinical guidance. This study proposed the design, development, and evaluation of the Smart Health Expert System (SHES), an AI-driven clinical decision support and preventive healthcare platform developed to improve access to preliminary medical assistance within the Kenyan context. The system was developed using the Agile Software Development Methodology to allow iterative planning, implementation, testing, and refinement. A full-stack architecture was adopted, utilizing React.js and TypeScript for the frontend, Django REST Framework for the backend, and PostgreSQL for database management, while the clinical reasoning component was powered by a forward-chaining rule-based expert system derived from Kenya National Clinical Guidelines and World Health Organization primary care protocols. SHES integrated multiple healthcare support functions, including AI symptom triage, chronic disease monitoring, drug interaction checking, laboratory result interpretation, mental wellness tracking, medication management, and secure patient medical profiling. System evaluation was carried out through unit testing, integration testing, functional testing, and user acceptance testing to determine reliability, usability, and clinical consistency. The findings indicated that the system accurately analyzed patient symptoms, generated appropriate health recommendations, securely managed sensitive medical records, and improved users' ability to monitor their health proactively. The study concludes that SHES has strong potential to enhance preventive healthcare delivery, reduce unnecessary hospital visits, and bridge the healthcare accessibility gap in Kenya through intelligent digital health support.

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LIST OF ABBREVIATIONS

AI	– Artificial Intelligence
API	– Application Programming Interface
BP	– Blood Pressure
CDSS	– Clinical Decision Support System
DFD	– Data Flow Diagram
DPA	– Data Protection Act
DRF	– Django REST Framework
ERD	– Entity Relationship Diagram
HbA1c	– Glycated Haemoglobin
HTTP	– Hypertext Transfer Protocol
ICT	– Information and Communications Technology
JNC	– Joint National Committee (on hypertension)
JWT	– JSON Web Token
KB	– Knowledge Base
KEML	– Kenya Essential Medicines List
ML	– Machine Learning
MOH	– Ministry of Health
NCD	– Non-Communicable Disease
OOP	– Object-Oriented Programming
REST	– Representational State Transfer
SHES	– Smart Health Expert System
SPA	– Single Page Application
SQL	– Structured Query Language
UAT	– User Acceptance Testing
UI	– User Interface

UUID – Universally Unique Identifier
WHO – World Health Organization

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CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

The intersection of artificial intelligence and healthcare informatics had, by the early 21st century, evolved from a theoretical concept into one of the most impactful domains

of applied computer science. Clinical decision support systems (CDSSs), which employed AI to provide clinicians and patients with patient-specific knowledge and intelligently filtered information relevant to specific health situations, represented a class of health information technology systems with demonstrated potential to improve diagnostic accuracy, reduce medical errors, and promote preventive health behaviours (Sutton et al., 2020). In the context of resource-constrained health systems such as Kenya's, the relevance of such systems was not merely academic but constituted a pragmatic response to a deepening public health crisis.

Kenya's public health system operated under extraordinary strain. According to the Kenya Health Workforce Report (Ministry of Health, 2021), the country's doctor-to-patient ratio stood at approximately 1:6,355, against the World Health Organization's recommended minimum threshold of 1:1,000 (WHO, 2020). This shortage was compounded by a deeply inequitable distribution of healthcare resources, with most trained medical personnel concentrated in urban centers such as Nairobi, Mombasa, and Kisumu, while large swathes of the rural population remained chronically underserved. The Kenya National Bureau of Statistics reported in 2022 that fewer than 50% of Kenyans in rural areas resided within five kilometers of a functioning healthcare facility, with this proportion dropping significantly in arid and semi-arid regions such as Turkana, Mandera, and Garissa counties.

Non-communicable diseases (NCDs) have presented a growing and largely under-managed burden within this already strained system. The Kenya STEPwise Survey for Non-Communicable Diseases Risk Factors (Ministry of Health, 2021) reported that hypertension affected approximately 24% of adults aged 18 years and above, while Type 2 diabetes mellitus was estimated to affect 2.4% of the adult population, with significant underdiagnosis suspected due to limited screening capacity. Managing these conditions required continuous patient monitoring, medication adherence, and regular clinical interaction — resources that the existing healthcare infrastructure was ill-equipped to provide at scale.

The digital health landscape offered a promising avenue for addressing these structural deficiencies. The proliferation of mobile telephone technology across sub-Saharan

Africa, which had achieved a mobile penetration rate of approximately 88% in Kenya by 2023 (Communications Authority of Kenya, 2023), created a technological infrastructure through which digital health interventions could reach populations previously excluded from formal health services. Early telemedicine initiatives, electronic health record systems, and mobile health applications had demonstrated measurable improvements in health seeking behavior, medication adherence, and disease management outcomes across various African contexts (Labrique et al., 2013; Mehl & Labrique, 2014).

Expert systems, a subcategory of artificial intelligence that encoded domain expertise into structured knowledge bases and applied formal reasoning mechanisms to produce context-specific outputs, had been applied in healthcare since the pioneering development of MYCIN at Stanford University in the 1970s — one of the first rule-based expert systems designed to identify bacterial infections and recommend antibiotic therapies (Shortliffe, 1976). Subsequent decades witnessed the evolution of such systems into modern CDSSs integrated into electronic health records, point-of-care devices, and standalone patient-facing applications. Contemporary systems such as Babylon Health's AI-powered symptom checker, Ada Health's probabilistic diagnostic assistant, and IBM Watson Health's oncology decision support tool demonstrated that AI-based clinical reasoning could achieve clinically meaningful and statistically reliable performance outcomes (Fraser et al., 2019).

However, globally available health technology platforms had been developed primarily for contexts characterized by well-resourced healthcare systems, high health literacy, and stable internet connectivity. Their clinical knowledge bases, disease profiles, treatment protocols, and user interface designs were calibrated for populations in North America, Europe, and high-income Asia — conditions that bore limited resemblance to the clinical realities, disease epidemiology, medication availability, and digital literacy levels prevalent in Kenya and sub-Saharan Africa more broadly. The critical gap that SHES was designed to fill was, therefore, not merely the absence of a health application, but the absence of a clinically grounded, contextually appropriate, and technically robust AI health platform designed from the ground up for Kenya's specific healthcare challenges, clinical guidelines, and population characteristics.

This project was undertaken against this backdrop of urgent need and demonstrated feasibility. SHES was conceived as a full-stack, AI-driven clinical decision support and preventive healthcare platform whose inference engine was explicitly grounded in the Kenya National Clinical Guidelines and WHO Primary Care Protocols, whose medication knowledge base was drawn from the Kenya Essential Medicines List, and whose system architecture was designed for deployment in conditions of variable internet connectivity and diverse digital literacy levels. The system represented not merely a software engineering exercise, but a deliberate and evidence-informed response to one of Kenya's most consequential and persistent public health challenges.

1.2 Statement of the Problem

Kenya's healthcare system confronted a multidimensional crisis whose complexity derived from the simultaneous convergence of structural inadequacies, demographic pressures, epidemiological transitions, and systemic inefficiencies. While individual dimensions of this crisis had been documented across numerous government reports, academic studies, and health sector analyses, their combined effect had not been comprehensively addressed through a unified digital health intervention prior to the development of SHES.

The first and most acute dimension of the problem was the persistent and worsening gap between the demand for primary healthcare consultation and the availability of trained healthcare professionals capable of providing it. As documented by the Ministry

of Health's Health Workforce Report (2021), Kenya possessed approximately 16,000 registered medical doctors serving a population of over 55 million people. The resulting doctor-to-patient ratio of approximately 1:3,437 at the national level — and considerably worse in rural counties — meant that the majority of Kenyan patients were unable to access timely medical consultation for even routine health complaints. Emergency rooms and outpatient departments in public referral hospitals regularly operated at two to three times their designed capacity, creating conditions in which genuinely urgent cases competed for clinical attention alongside non-emergency presentations that could reasonably have been managed through self-care guidance or telemedicine consultation (Kenya Health Service Delivery Report, 2022).

The second dimension of the problem was the absence of accessible, reliable, and contextually appropriate tools for preliminary symptom assessment and triage. Kenyans seeking to understand the clinical significance of their symptoms prior to seeking formal care had access to global health platforms such as WebMD and NHS Choices, but these platforms' content was calibrated for disease profiles, treatment options, medication availability, and healthcare pathways that were fundamentally different from Kenya's. Critical diseases in the Kenyan context — including malaria, typhoid fever, anaemia, and schistosomiasis — were either absent from or insufficiently represented in these platforms, while their drug interaction databases and prescribing guidance failed to account for the medications available through Kenya's health system. The consequence of this gap was that patients either delayed seeking care due to uncertainty about their condition's severity, or presented to overcrowded healthcare facilities unnecessarily, contributing to the congestion that degraded care quality for all patients.

The third dimension of the problem was the poor state of chronic disease management at the individual patient level. Hypertension, Type 2 diabetes mellitus, and other non-communicable diseases require ongoing, patient-engaged self-monitoring as an essential complement to clinical management. Yet in the absence of accessible tools for recording, interpreting, and acting on blood pressure and blood glucose readings, the majority of Kenyan patients managed these conditions reactively — presenting to health facilities only when symptoms became acute — rather than proactively

monitoring trends and adjusting behavior accordingly. The resulting pattern of episodic, crisis-driven care contributed to high rates of preventable complications, including hypertensive heart disease, diabetic nephropathy, and lower limb amputations, which placed further strain on already stretched hospital resources (Kenya Non-Communicable Disease Alliance, 2021).

The fourth dimension of the problem was the systemic neglect of medication adherence as a public health intervention. The World Health Organization estimated that only approximately 50% of patients with chronic diseases in developing countries adhered to prescribed medication regimens (WHO, 2019). In Kenya, this problem was exacerbated by low health literacy, complex polypharmacy regimens for patients managing multiple chronic conditions, and the absence of accessible tools for tracking medication schedules, understanding drug interactions, and receiving timely reminders. The consequences of poor adherence ranged from subtherapeutic drug levels and treatment failure to toxicity from inconsistent dosing patterns, all of which contributed to preventable morbidity and mortality.

The fifth and final dimension of the problem was the near-total absence of integrated mental health support within Kenya's digital health landscape. Kenya's mental health system was severely under-resourced, with fewer than 100 psychiatrists serving a population of over 55 million and a treatment gap for common mental disorders estimated at over 85% (Africa Mental Health Research and Training Foundation, 2020). Depression, anxiety, and mood disorders contributed significantly to the overall burden of disease yet remained largely invisible within both clinical practice and digital health applications. The absence of accessible, evidence-based mood tracking and mental health support tools constituted both a clinical gap and a missed opportunity for preventive intervention at the population level.

SHES was developed in response to this five-dimensional problem. The central research question addressed by the project was: could a locally grounded, AI-driven clinical decision support platform, designed in explicit alignment with Kenya's clinical guidelines, disease epidemiology, and healthcare infrastructure, meaningfully improve

access to preliminary clinical guidance, chronic disease self-management, and preventive health behaviors for Kenyan patients?

1.3 Proposed Solution

The proposed solution to the identified healthcare access problem was the design, development, and deployment of SHES — the Smart Health Expert System — a full-stack, AI-driven clinical decision support and preventive healthcare platform built explicitly to serve the Kenyan healthcare context. SHES was conceived not as an incremental improvement to existing global health applications but as an integrated, purpose-built platform that addressed the identified problem dimensions through a coherent set of functionally specialized but technically interconnected modules.

The core intelligence of SHES resided in its forward-chaining rule-based inference engine, which encoded clinical decision logic drawn directly from the Kenya National Clinical Guidelines and WHO Primary Care Protocols. The engine processed patient-reported symptom inputs — including symptom names, severity ratings on a scale of one to ten, and duration in days — and executed a multi-step reasoning process that scanned for twelve emergency red flag patterns, matched the presented symptom profile against a knowledge base of fourteen clinical conditions prevalent in the Kenyan

context, applied severity and duration escalation rules, and produced a structured triage output comprising an urgency level classification, a plain-language clinical recommendation, matched conditions with confidence ratios, and personalised home-care guidance. The four urgency levels — Emergency, Doctor Visit, Self-Care, and Undetermined — were calibrated to guide patients toward appropriate care pathways without inducing unnecessary alarm or providing false reassurance.

Complementing the triage engine, SHES incorporated a Chronic Disease Tracking module that enabled patients to log blood glucose readings across multiple measurement contexts (fasting, post-meal, random, and bedtime) and blood pressure readings with automatic classification against WHO glycaemic targets and JNC-8 blood pressure categories respectively. Longitudinal trend visualizations built using the Recharts library enabled patients and clinicians to monitor health trajectories over time, supporting the proactive, data-driven approach to chronic disease management that was absent from the existing care ecosystem.

A Drug Interaction Checker grounded in Kenya's Essential Medicines List enabled patients on polypharmacy regimens to identify potentially dangerous drug interactions before they resulted in clinical harm. The Mental Health Module provided an accessible, evidence-based mood tracking and coping strategy recommendation tool that addressed the documented treatment gap in Kenya's mental health landscape. A Laboratory Results Interpreter translated laboratory values for nine common diagnostic tests into plain-language explanations and actionable recommendations, removing a significant barrier to patient understanding of their own health data. The Patient Medical Profile module maintained comprehensive longitudinal health records including blood group, documented allergies, and chronic condition history, while the role-based authentication system ensured data security and appropriate access control in compliance with the Kenya Data Protection Act, 2019.

The proposed solution was thus distinguished from existing alternatives in four fundamental respects: clinical grounding in Kenya-specific guidelines and disease profiles; technical depth in terms of security architecture, test coverage, and production-readiness; breadth of functional coverage spanning triage, chronic disease management,

mental health, and medication management within a single integrated platform; and contextual appropriateness in terms of language, user interface design, and clinical logic calibrated for the realities of healthcare in Kenya.

1.4 Objectives

1.4.1 Main Objective

The main objective of this project was to design, develop, and evaluate an AI-driven clinical decision support and preventive healthcare platform — SHES — capable of providing Kenyan patients and healthcare providers with accessible, evidence-based, and clinically grounded health guidance to reduce delayed diagnosis, improve chronic disease self-management, and promote preventive health behaviors.

1.4.2 Specific Objectives

The specific objectives of the study were as follows:

1. To analyze the existing challenges in Kenya's primary healthcare consultation system and identify the functional requirements for an AI-driven clinical decision support platform capable of addressing those challenges.

2. To design a forward-chaining rule-based clinical inference engine grounded in the Kenya National Clinical Guidelines and WHO Primary Care Protocols, capable of processing patient-reported symptoms and generating clinically appropriate urgency classifications and recommendations.
3. To develop and implement a full-stack web-based health management platform comprising symptom triage, chronic disease tracking, medication management, drug interaction checking, laboratory result interpretation, mental health support, and patient medical history management modules.
4. To integrate a role-based, JWT-secured authentication system with field-level encryption compliant with the Kenya Data Protection Act, 2019, ensuring that patient health data was protected from unauthorized access.
5. To evaluate the functional correctness, security robustness, and clinical reliability of the developed system through a comprehensive testing programme comprising unit, integration, functional, and user acceptance testing.

1.5 Significance of the Study

The significance of this project extended across academic, clinical, and policy dimensions, each of which warranted explicit articulation.

From an academic and computer science perspective, this project contributed a working implementation of a context-specific, clinically grounded expert system to the body of health informatics research in Kenya. While the international literature contained numerous studies of AI-based clinical decision support systems, empirical work grounded in Kenya's specific clinical guidelines, disease epidemiology, and healthcare infrastructure was sparse. The SHES project demonstrated that a production-grade, full-stack health AI platform could be designed and built within the resource constraints of an undergraduate research project, and the detailed documentation of the design decisions, technical architecture, inference engine logic, and testing outcomes contributed a replicable methodological template for future health informatics research in the Kenyan and East African context.

From a clinical and public health perspective, the practical significance of the project lay in its potential to extend the effective reach of Kenya's overstretched clinical workforce. By providing patients with access to a reliable, evidence-based preliminary triage tool, SHES created a mechanism through which genuinely urgent cases could be prioritized for immediate clinical attention while non-urgent presentations received

appropriate self-care guidance without consuming scarce clinical resources. The chronic disease management modules held particular significance given the growing burden of hypertension and diabetes in Kenya, offering patients the tools for continuous self-monitoring and data-driven engagement with their own health that existing clinical systems failed to provide.

For rural and underserved communities — including pastoralist populations in arid counties, smallholder farming communities in remote highland areas, and urban informal settlement residents — the significance of a mobile-accessible, internet-light health platform was particularly acute. These populations experienced the most severe manifestations of Kenya's healthcare access gap and had historically been the least served by digital health innovations. SHES was designed with these populations explicitly in mind, with a lightweight single-page application architecture, responsive design, and plain-language clinical outputs calibrated for users with varying levels of health and digital literacy.

From a policy perspective, the project contributed practical evidence for the viability of AI-assisted primary health care as a component of Kenya's Universal Health Coverage implementation strategy. The Kenya Health Policy 2014–2030 and the Big Four Agenda both identified accessible, affordable primary healthcare as a national priority. SHES provided a concrete demonstration of how AI technology, grounded in Kenya's own clinical standards and integrated with the country's existing medicines list and clinical guidelines, could be deployed to support this policy objective.

1.6 Scope of the Study

The scope of this project encompassed the design, development, testing, and documentation of the SHES platform as a web-based application optimized for both desktop and mobile browser access. The system was developed as a locally hosted, cloud-ready application, with full deployment to a production cloud environment planned as a post-project extension.

In terms of clinical scope, the inference engine covered fourteen clinical conditions identified as highly prevalent and clinically significant within Kenya's primary healthcare context: Common Cold, Influenza, Malaria, Typhoid Fever, Gastroenteritis, Urinary Tract Infection, Hypertension, Type 2 Diabetes Mellitus, Tension Headache, Anaemia, Skin Infection, Anxiety Disorder, Lower Back Pain, and Asthma. The emergency red flag module covered twelve life-threatening symptom patterns including chest pain, stroke signs, suicidal ideation, severe allergic reaction, seizure, and suspected meningitis. The drug interaction database encompassed medications from the Kenya Essential Medicines List with documented interaction pairs. The laboratory results interpreter covered nine common diagnostic tests including haemoglobin, HbA1c, fasting blood glucose, total cholesterol, serum creatinine, white blood cell count, platelet count, LDL cholesterol, and HDL cholesterol.

The scope explicitly excluded diagnostic imaging interpretation, genomic data analysis, prescription generation, real-time telemedicine video consultation, integration with

Kenya's national health information system (KHIS), and direct electronic health record integration with hospital management systems. These exclusions reflected both the technical constraints of an undergraduate project timeline and the regulatory boundaries within which a non-licensed health technology platform appropriately operated.

The target user population encompassed adult patients aged eighteen years and above seeking preliminary health guidance, registered medical doctors and healthcare administrators using the platform's clinical management interfaces, and preventive healthcare seekers engaged in proactive health monitoring. Pediatric-specific clinical logic was not incorporated within the current version of the system, representing an area for future extension.

1.7 Assumptions

The development and evaluation of SHES proceeded based on several foundational assumptions that were explicitly acknowledged:

- Users were assumed to possess sufficient digital literacy to navigate a web-based application, including the ability to register an account, input symptom data, and interpret system-generated outputs in plain English.
- Users were assumed to have access to a device capable of running a modern web browser — including smartphones, tablets, and personal computers — with an internet connection of sufficient stability to load a single-page web application.
- The clinical knowledge encoded in the inference engine was assumed to remain valid and authoritative for the duration of the project, with the understanding that clinical guidelines were periodically updated and that the knowledge base would require ongoing maintenance to remain current.
- Users were assumed to report symptoms honestly and to the best of their ability, with the understanding that the quality of the system's triage output was directly dependent on the accuracy and completeness of the symptom information provided.
- The system was not assumed to replace professional medical diagnosis or clinical judgment. All triage outputs were explicitly framed as advisory and preliminary, and users were directed to seek professional clinical care for definitive diagnosis and treatment.

1.8 Limitations and Delimitations

1.8.1 Limitations

The study was subject to several limitations that were acknowledged as potential constraints on its generalizability and completeness:

The inference engine employed a forward-chaining rule-based architecture rather than a probabilistic machine learning model. While this design choice enabled transparent, interpretable clinical reasoning and simplified knowledge base maintenance, it did not produce the probabilistic confidence scores that Bayesian or neural network-based models could generate. The system's accuracy was consequently bounded by the completeness and correctness of the manually curated clinical rule set.

The knowledge base covered fourteen clinical conditions and twelve emergency red flags, which, while representative of common primary care presentations in Kenya, did not encompass the full spectrum of conditions that a patient might present with. The system's response to symptom profiles associated with conditions outside this set was an 'Undetermined' urgency classification — a safe but informationally limited outcome.

User acceptance testing was conducted with a convenience sample of participants recruited from the researcher's academic and professional network, which may not have been fully representative of the diverse population of intended users, particularly older adults and users with limited digital literacy.

1.8.2 Delimitations

The project was deliberately delimited to exclude real-time telemedicine consultation, integration with external electronic health record systems, and generation of prescriptions or formal clinical documentation. These exclusions were not technical limitations, but deliberate scope boundaries established to maintain the project's focus, manage development timelines, and ensure the system operated appropriately within the regulatory framework applicable to non-licensed health technology platforms in Kenya.

1.9 Definition of Terms

The following terms were used throughout this report with the specific meanings defined below:

Artificial Intelligence (AI): The simulation of human cognitive processes — including reasoning, learning, and problem-solving — by computer systems, applied in this project through a rule-based inference engine that performed clinical reasoning on patient-reported symptom data.

Clinical Decision Support System (CDSS): A health information technology application designed to assist clinicians and patients in making evidence-based health decisions by providing context-relevant, patient-specific clinical knowledge at the point of care.

Expert System: A computer program that emulates the decision-making ability of a human expert within a defined domain by encoding domain knowledge in a structured knowledge base and applying formal inference mechanisms to derive conclusions.

Forward Chaining: A data-driven inference strategy in which the reasoning process proceeded from known facts (patient-reported symptoms) to derived conclusions (clinical urgency classification) by applying IF-THEN rules stored in the knowledge base.

Knowledge Base: The structured repository of encoded clinical rules, condition profiles, emergency red flag definitions, and drug interaction records that constituted the domain knowledge upon which the SHES inference engine operated.

Inference Engine: The computational component of an expert system responsible for applying the rules stored in the knowledge base to input data to derive conclusions. In

SHES, this was implemented as the Python-based Inference Engine class within the triage application module.

Triage: The clinical process of sorting patients according to the urgency of their medical condition and directing them toward the appropriate level of care. In the SHES context, triage referred to the automated classification of a patient's reported symptom profile into one of four urgency categories.

Non-Communicable Disease (NCD): A chronic disease that was not transmissible from person to person, including hypertension, Type 2 diabetes mellitus, cardiovascular disease, and chronic respiratory conditions, all of which were addressed within SHES's clinical scope.

JWT (JSON Web Token): A compact, URL-safe means of representing claims between two parties, used in SHES as the mechanism for stateless user authentication and authorization across the REST API.

Field-Level Encryption: A data security technique in which individual sensitive data fields within a database record — rather than the entire database — were encrypted using a symmetric encryption algorithm, reducing the risk of exposure in the event of a breach. SHES employed Fernet symmetric encryption (AES-128-CBC) for sensitive health data fields.

Kenya Essential Medicines List (KEML): The official list of medications maintained by the Kenya Ministry of Health that satisfied the priority healthcare needs of the population, selected with due regard to disease prevalence, efficacy, safety, and comparative cost-effectiveness. The KEML constituted the drug catalogue for SHES's Medication Management and Drug Interaction modules.

END OF CHAPTER ONE

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter presented a critical examination of the existing body of knowledge pertaining to digital health systems, artificial intelligence in clinical decision support, expert system architectures, and health informatics applications in sub-Saharan Africa. The review was organized to progress from broad theoretical frameworks to specific related systems, culminating in a comparative analysis that identified the research gap that the development of SHES was designed to address. The chapter drew upon peer-reviewed journal articles, government health reports, technical documentation from health technology platforms, and authoritative guidelines from national and international health bodies. All sources were evaluated for relevance, methodological rigor, and applicability to the Kenyan healthcare context.

2.2 Digital Health Systems and Health Informatics

2.2.1 Definition and Scope of Health Informatics

Health informatics, defined by the American Medical Informatics Association as the interdisciplinary field that studies and pursues the effective use of biomedical data, information, and knowledge for scientific inquiry, problem-solving, and decision-making (AMIA, 2021), had by the second decade of the 21st century emerged as one of the most consequential intersections of computer science and public health. The discipline encompassed electronic health records, clinical decision support systems, telemedicine platforms, health data analytics, and patient-facing health applications — each of which represented a distinct but interrelated mechanism through which computational tools were applied to improve health outcomes.

Shortliffe and Cimino (2014) characterized the core objective of health informatics as the management of health information to support clinical practice, education, and research. This framing was particularly relevant to the development of SHES, which was concerned with transforming raw patient-reported symptom data into structured clinical knowledge that could guide appropriate health-seeking behavior — a task that demanded both computational sophistication and deep clinical grounding.

The World Health Organization's Global Strategy on Digital Health 2020–2025 identified digital health interventions as a critical enabler of Universal Health Coverage, noting that technologies including AI-powered clinical tools, mobile health applications, and telemedicine platforms had the potential to dramatically extend the reach of health services to underserved populations (WHO, 2021). This strategic framing provided an important policy context for SHES, positioning it not merely as a software engineering project but as a contribution to a globally recognized public health imperative.

2.2.2 Evolution of Clinical Decision Support Systems

The intellectual lineage of clinical decision support systems traced back to the pioneering work of Edward Shortliffe at Stanford University in the 1970s. The MYCIN system, developed between 1972 and 1980, applied a backward-chaining rule-based inference mechanism to identify the causative organisms of bacterial infections and recommend appropriate antibiotic therapies (Shortliffe, 1976). MYCIN's significance extended beyond its clinical domain: it demonstrated that AI-encoded clinical expertise could achieve diagnostic accuracy comparable to that of experienced specialists — a finding that fundamentally transformed the research agenda of medical informatics for the subsequent four decades.

Subsequent pioneering systems expanded the clinical scope of rule-based expert systems. INTERNIST-1, developed at the University of Pittsburgh in the 1980s, employed a causal-associational model to perform differential diagnosis across a broad range of internal medicine conditions (Miller et al., 1982). CADUCEUS, its successor, introduced temporal reasoning capabilities that enabled the system to reason about disease progression over time — a feature whose conceptual influence could be observed in the duration-based escalation logic implemented in SHES's inference engine.

The evolution of CDSSs from standalone diagnostic tools to integrated components of electronic health record systems accelerated significantly in the 1990s and 2000s. Systems such as Arden Syntax, Heckerling's rule-based pneumonia diagnosis system, and the integrated CDSSs embedded within commercial electronic health record platforms such as Epic and Cerner demonstrated that AI-assisted clinical reasoning could be deployed at scale within complex clinical information environments (Berner, 2009). These systems shared a common architectural pattern — a structured knowledge base encoding clinical rules, an inference mechanism applying those rules to patient data, and a presentation layer communicating outputs to clinicians — a pattern that informed the architectural design of SHES.

More recent developments in the field have introduced probabilistic and machine learning approaches that supplemented or replaced rule-based reasoning. Deep learning models trained on large clinical datasets had demonstrated performance comparable to specialist clinicians in domains including dermatological image classification, diabetic retinopathy screening, and radiological interpretation (Topol, 2019). However, the deployment of such models in resource-constrained settings faced significant challenges: they required large, high-quality training datasets that were often unavailable in low-income country contexts; their reasoning processes were frequently opaque, raising concerns about clinical accountability and patient trust; and their maintenance and updating required machine learning expertise that was scarce in many healthcare settings. These considerations provided a strong justification for SHES's adoption of a transparent, interpretable, and manually maintainable rule-based inference architecture.

2.2.3 Digital Health in Sub-Saharan Africa

The application of digital health technologies in sub-Saharan Africa had been the subject of growing research attention since the mid-2000s. Early interventions focused primarily on mobile-based health communication, appointment reminder systems, and community health worker support tools — low-complexity applications that leveraged the region's rapidly expanding mobile telephone infrastructure (Labrique et al., 2013). Systematic reviews of these early interventions consistently demonstrated improvements in health-seeking behavior, medication adherence, and vaccination uptake, establishing compelling evidence base for the potential of digital tools to improve health outcomes in low-resource settings (Free et al., 2013).

More sophisticated AI-based applications had begun to emerge within the African health technology ecosystem, with notable examples including mPedigree's drug authentication system in Ghana, the Peek Vision smartphone-based ophthalmological assessment tool deployed in Kenya and Rwanda, and KEMR, Kenya's electronic medical record system deployed across public health facilities (Ministry of Health Kenya, 2018). These systems demonstrated that complex health technology solutions could be designed, implemented, and sustained within African health systems, though the majority remained narrowly focused on specific clinical domains rather than providing comprehensive primary care support.

A systematic review by Mehl and Labrique (2014) identified fourteen distinct categories of digital health intervention with evidence of effectiveness in low- and middle-income countries, including decision support for healthcare workers, client-targeted reminder systems, and electronic health records. The review noted, however, that few existing systems integrated multiple intervention types within a single platform — an observation that underscored the innovation represented by SHES's multi-module architecture.

The Kenya-specific digital health landscape had been shaped by several government-led initiatives. The Kenya Health Policy 2014–2030 articulated a vision of a transformed health system leveraging information and communications technology to improve service delivery, while the Kenya National eHealth Strategy provided a roadmap for the integration of digital tools across all levels of the health system (Ministry of Health Kenya, 2016). SHES was developed in explicit alignment with these policy frameworks, contributing a practical implementation that addressed the AI-powered clinical decision support component of the eHealth strategy that had remained underdeveloped in the published literature.

2.3 Review of Related Systems

The following subsections presented a critical analysis of three commercially deployed health technology platforms that were most closely related to SHES in terms of functional scope and target use cases. Each system was examined with respect to its clinical approach, technical architecture, target population, strengths, and weaknesses relative to the Kenyan healthcare context.

2.3.1 Babylon Health

Babylon Health was a digital health company founded in the United Kingdom in 2013, offering an AI-powered symptom assessment service alongside telemedicine consultation, chronic disease management tools, and electronic prescribing. The platform was deployed at scale in the United Kingdom through a partnership with the National Health Service and expanded internationally, including a significant deployment in Rwanda — one of the few African countries where the platform had been evaluated in a low-income context (Babylon Health, 2020).

The platform's symptom checker employed a combination of natural language processing and a probabilistic Bayesian reasoning model trained on large volumes of clinical data from UK and international healthcare settings. Published evaluations of Babylon's symptom assessment capability reported that the system's triage recommendations matched the recommendations of human clinicians in 81% of cases in a study conducted using clinical vignettes representative of the UK primary care context (Fraser et al., 2019). However, this performance evaluation was conducted against a UK clinical standard, raising significant questions about the system's reliability when applied to the disease profiles, medication availability, and clinical pathways characteristic of the Kenyan context.

A critical analysis of Babylon Health's relevance to Kenya identified several substantive limitations. First, the platform's clinical knowledge base was calibrated for the UK National Health Service's disease burden, treatment protocols, and referral pathways — none of which were directly applicable to Kenya's primary care system. Conditions of high clinical significance in Kenya — including malaria, typhoid fever, sickle cell disease, and schistosomiasis — were either absent from or inadequately represented in the system's condition library. Second, Babylon's business model relied on a freemium subscription structure that placed its most clinically sophisticated features behind a paywall, limiting accessibility for the majority of the Kenyan population who lacked the disposable income to sustain a health application subscription. Third, the platform's AI model was a black-box probabilistic system whose reasoning processes were not auditable by users or clinicians, raising concerns about clinical accountability and patient trust — particularly in settings where health literacy was variable and trust in technology-mediated health services was still being established.

Despite these limitations, Babylon Health made several important conceptual contributions to the field that informed the design of SHES. Its integration of symptom assessment, chronic disease management, and telemedicine within a single platform demonstrated the viability of a comprehensive digital primary care approach. Its use of clinician-reviewed clinical vignettes as an evaluation methodology provided a rigorous benchmark approach that informed the design of SHES's functional testing programme.

2.3.2 Ada Health

Ada Health was a Berlin-based digital health company founded in 2011, offering an AI-powered symptom assessment application that employed a probabilistic differential diagnosis model to generate ranked lists of conditions consistent with a user's reported symptom profile. The platform had been deployed in over 130 countries and had reportedly conducted over 20 million symptom assessments as of 2022 (Ada Health, 2022). It was notable for its multilingual capability and its stated commitment to global health equity, positioning itself as a digital health tool with relevance to low-income country settings.

Ada's clinical approach employed a Bayesian inference model that estimated the conditional probability of each condition in its database given the observed symptom profile, generating ranked differential diagnosis lists that were updated dynamically as users added symptom information. The system's probabilistic architecture enabled it to express uncertainty in its assessments — a clinically important feature that avoided the false certainty that could accompany more deterministic diagnostic tools. A validation study published in *Nature Digital Medicine* reported that Ada's symptom checker correctly included the true diagnosis in its top five differential diagnoses in 80% of cases evaluated against a set of 1,000 clinical vignettes drawn from the British Medical Journal case database (Raza et al., 2020).

Critical analysis of Ada Health's applicability to the Kenyan context revealed several significant limitations. The system's probabilistic model was trained on clinical data predominantly drawn from European and North American healthcare settings, and its condition library, while broad, did not adequately represent the tropical disease burden that characterized Kenya's primary care morbidity profile. The platform provided symptom assessment but did not offer integrated chronic disease monitoring, medication management, drug interaction checking, or laboratory results interpretation — meaning that users required multiple separate applications to address the full range of health management needs that SHES addressed within a single integrated platform. Additionally, Ada's user interface, while multilingual, was not localised for Kenya's specific healthcare pathways, emergency contact systems, or medication landscape — limiting its utility as a point-of-care triage tool.

From a technical perspective, Ada's probabilistic architecture, while offering theoretical advantages in uncertainty quantification, presented practical limitations in terms of transparency and maintainability in low-resource settings. The system's model parameters were not publicly documented, and the process of updating the knowledge base to reflect new clinical evidence or local guidelines was not accessible to healthcare administrators or clinical informatics teams in Kenya. SHES's JSON-based knowledge base architecture, by contrast, was explicitly designed for ease of clinical content updates by local clinical teams without requiring machine learning expertise.

2.3.3 WebMD Symptom Checker

WebMD was one of the longest-established and most widely accessed health information platforms in the world, with an estimated 80 million unique monthly visitors globally as of 2022 (WebMD, 2022). Its Symptom Checker feature enables

users to select symptoms from a predefined list and receive a list of possible conditions consistent with those symptoms, alongside links to further health information. The platform had existed in various forms since the late 1990s and represented the most widely used consumer-facing symptom assessment tool globally.

WebMD's symptom checker employed a rule-based matching algorithm that compared the user's selected symptoms against a database of conditions and returned a list of possible diagnoses ranked by symptom-condition match frequency. The platform's strength lay in the breadth of its health information library, the accessibility of its interface, and the breadth of its disease coverage within the US clinical context. A consumer survey conducted by WebMD reported high levels of user satisfaction with the platform's symptom information, with most respondents indicating that the platform influenced their decision about whether to seek medical care (WebMD Consumer Survey, 2021).

However, WebMD's applicability to the Kenyan healthcare context was substantially limited by several structural characteristics of the platform. Its disease database was constructed around the US healthcare system's disease burden, treatment protocols, and clinical pathways, with limited representation of tropical and endemic diseases prevalent in Kenya. The platform's drug information module was calibrated for medications available within the US pharmaceutical market, with minimal representation of the Kenya Essential Medicines List or the treatment protocols used in Kenya's public health facilities. The symptom checker provided no emergency triage logic — users presenting symptoms consistent with life-threatening conditions received the same ranked condition list as those presenting mild, self-limiting complaints, with no mechanism for flagging emergency presentations and directing users toward immediate care.

From a technical perspective, WebMD's symptom checker was a consumer information tool rather than a clinical decision support system. It provided ranked condition lists without urgency classifications, clinical recommendations, or personalized home-care guidance — outputs that SHES generated through its structured inference engine. The platform was also ad-supported, meaning that users' health data and browsing behavior contributed to a commercial advertising model, raising data privacy concerns that were particularly significant in the context of sensitive health information. SHES, by contrast, was designed as a non-commercial, privacy-focused platform with field-level encryption and explicit compliance with the Kenya Data Protection Act, 2019.

2.4 Comparative Analysis of Related Systems

Table 2.1 below presented a systematic comparison of the three reviewed systems against SHES across twelve evaluative dimensions. The comparison was constructed to highlight the specific features and design choices that distinguished SHES from existing platforms and justified its development as an original contribution to the health informatics landscape.

Table 2.1: Comparative Analysis of Related Health Technology Platforms

Feature / Criterion	Babylon Health	Ada Health	WebMD Symptom Checker	SHES (Proposed)
Clinical Knowledge Grounding	UK/US clinical protocols; no Kenya guidelines	Global probabilistic model; non-contextualized	US-based content; limited African relevance	Kenya National Clinical Guidelines + WHO Primary Care Protocols
Disease Coverage	Primarily high-income disease profiles	Wide but generic; lacks tropical disease depth	Limited tropical disease coverage	14 conditions incl. Malaria, Typhoid, Anaemia, NCD-focused
Inference Mechanism	Probabilistic NLP + ML model	Bayesian probabilistic reasoning	Rule-based symptom matching	Forward-chaining rule-based engine; transparent & auditable
Emergency Red Flag Detection	Present but calibrated for UK system	Present; generalized	Limited; no formal escalation logic	12 emergency red flags with immediate EMERGENCY routing
Chronic Disease Monitoring	Not available as a core feature	Limited; no longitudinal tracking	Absent	Full BP + glucose tracking with WHO/JNC-8 auto-classification and trend charts
Drug Interaction Checker	Available via separate service	Absent	Present but US drug catalogue	Integrated; grounded in Kenya Essential Medicines List
Mental Health Module	Offered in premium subscription	Basic mental health symptoms only	Absent	Full mood tracking + coping strategies + 14-day trend analysis
Lab Results Interpreter	Absent	Absent	Absent	9 common tests; plain-language interpretation per MOH

				reference ranges
Data Privacy Compliance	GDPR compliant; UK/EU focus	GDPR compliant	HIPAA compliant; US focus	Kenya Data Protection Act 2019; Fernet field-level encryption
Open Source / Local Adaptability	Proprietary; closed API	Proprietary; limited localization	Proprietary	Open source; knowledge base extensible via JSON; locally deployable
Cost / Accessibility	Freemium; full features behind subscription	Freemium model	Free but ad-supported; US-centric	Designed for free public access; open deployment model

Source: Researcher's own compilation based on platform documentation and published evaluations (2026)

The comparative analysis in Table 2.1 revealed a consistent and significant pattern: existing platforms, while technically sophisticated and clinically validated within their target contexts, failed to address the specific combination of clinical, contextual, and functional requirements that characterized Kenya's healthcare landscape. None of the reviewed systems provided clinical knowledge grounded in Kenya's National Clinical Guidelines; none offered an integrated platform combining triage, chronic disease monitoring, mental health support, drug interaction checking, and laboratory results interpretation; and none were designed for deployment and maintenance within Kenya's health information infrastructure.

This comparative analysis confirmed the existence of a substantive and unaddressed research and implementation gap that SHES was designed to fill — not through incremental improvement to existing systems but through the ground-up design of a platform that was Kenyan in its clinical grounding, comprehensive in its functional scope, transparent in its reasoning processes, and practically deployable within the resource constraints of Kenya's healthcare system.

2.5 Theoretical Frameworks Informing SHES

2.5.1 Expert System Theory and Forward Chaining

The theoretical foundation of SHES's clinical inference engine was grounded in the classical theory of rule-based expert systems as articulated by Hayes-Roth, Waterman, and Lenat (1983) in the foundational text 'Building Expert Systems.' This theoretical

framework described an expert system as comprising three essential components: a knowledge base encoding domain expertise in the form of IF-THEN production rules, a working memory containing the current state of known facts, and an inference engine applying the rules in the knowledge base to the facts in working memory to derive new conclusions.

SHES implemented a forward-chaining inference strategy, which Hayes-Roth et al. (1983) characterized as a data-driven approach in which the reasoning process was initiated from the available input data — in SHES's case, the patient's reported symptom profile — and proceeded through rule application until a conclusion was reached. This approach contrasted with backward chaining, which began from a hypothesized conclusion and sought evidence to confirm or refute it. Forward chaining was selected for SHES because it aligned naturally with the clinical triage process: rather than testing a preformed diagnostic hypothesis, the system began with what the patient knew — their symptoms — and derived the clinically appropriate response from that starting point.

The theoretical framework also informed the design of SHES's knowledge of representation architecture. Clancey (1983), in a seminal critique of MYCIN, argued that expert systems whose knowledge was encoded as opaque, monolithic rule sets were difficult to maintain, validate, and extend — a critique that motivated SHES's adoption of a modular, JSON-based knowledge base that separated clinical rules, condition profiles, and recommendations into distinct, independently maintainable files. This architectural choice also aligned with the principle of knowledge base transparency articulated by Shortliffe and Buchanan (1975), which held that clinical decision support systems should make their reasoning processes accessible to clinicians and users to build the trust necessary for clinical adoption.

2.5.2 Health Belief Model

The behavioral dimension of SHES's design was informed by the Health Belief Model (HBM), originally developed by Rosenstock (1966) and subsequently refined by Becker (1974). The HBM proposed that health-related behavior was determined by a set of perceived constructs including perceived susceptibility to a health threat, perceived severity of the threat, perceived benefits of the health action, perceived barriers to the action, and cues to action that triggered health-seeking behavior.

The relevance of the HBM to SHES's design was twofold. First, the system's urgency classification outputs — Emergency, Doctor Visit, Self-Care, Undetermined — were explicitly calibrated to function as cues to action that communicated perceived severity in a manner that motivated appropriate health-seeking responses without inducing excessive anxiety or complacency. Plain-language clinical recommendations and home-care tips addressed both the perceived benefits of appropriate action and the perceived barriers of uncertainty about what to do. Second, the chronic disease tracking modules were designed as ongoing tools for reinforcing perceived susceptibility — by visualizing longitudinal blood pressure and glucose trends, the system made the health

consequences of NCD management visible and personally relevant in a manner that encouraged sustained engagement with monitoring and care-seeking behavior.

2.5.3 Human-Computer Interaction in Health Technology

The user interface design principles applied in SHES were informed by the established literature on human-computer interaction in health technology. Cengiz and Ozbay (2022) identified five key usability dimensions that were critical to patient-facing health applications in low-resource settings: learnability, efficiency, memorability, error prevention, and user satisfaction. The authors noted that health applications failing on any of these dimensions risked abandonment by users — a particularly consequential failure in the context of chronic disease management tools requiring sustained daily engagement.

Norman's (2013) principles of human-centred design — particularly the concepts of affordance, feedback, and constraints — were applied in the design of SHES's symptom input interface, which used explicit severity rating sliders, duration input fields, and clear labelling to guide users through the symptom assessment process while minimizing the risk of input errors that would degrade the quality of the inference engine's output. The system's plain-language triage output design was informed by Paasche-Orlow and Wolf's (2007) health literacy framework, which argued that health communications should be calibrated to the reading level and health knowledge base of their target audience rather than the domain expertise of their developers.

2.6 The Kenyan Healthcare Context: Evidence from the Literature

A substantial body of peer-reviewed literature documented the specific dimensions of Kenya's healthcare access challenge that informed the problem definition and system design of SHES. Barasa et al. (2018), in a comprehensive analysis of Kenya's health financing landscape, documented that out-of-pocket health expenditure accounted for approximately 28% of total health spending in Kenya, with catastrophic health expenditure — defined as health spending exceeding 40% of household non-food income — affecting approximately 1.5% of Kenyan households annually. These financial barriers to formal healthcare access reinforced the public health case for digital tools capable of reducing the frequency of unnecessary formal healthcare contacts.

Muchiri and Kimani (2021) examined the pattern of healthcare utilization in Kenya's informal settlements and rural areas, documenting that the most common reason cited by respondents for not seeking formal care for symptomatic illness was uncertainty about whether their symptoms warranted a clinical consultation — a finding that directly validated the use case for SHES's symptom triage module. The study reported that approximately 64% of respondents who had experienced symptoms they considered potentially concerning in the preceding three months had not sought formal care, citing cost, distance, and uncertainty as the primary barriers.

Muthoni et al. (2020) conducted a systematic review of mobile health interventions for chronic disease management in sub-Saharan Africa, finding that digital reminders and self-monitoring tools were associated with statistically significant improvements in medication adherence for hypertension and diabetes management. The review identified the absence of integrated platforms combining medication reminders, vital sign monitoring, and clinical guidance as a critical gap in the existing evidence base — a gap that SHES's multi-module architecture was designed to address.

Africa Mental Health Research and Training Foundation (2020) documented that Kenya's mental health treatment gap — the proportion of individuals with a diagnosable mental disorder not receiving treatment — exceeded 85%, with mental health conditions accounting for an estimated 14% of the total burden of disease. These statistics provided the empirical foundation for SHES's inclusion of a Mental Health Module as a core component of the platform, rather than an optional or supplementary feature.

Waithaka et al. (2019) evaluated the quality of clinical decision support provided by existing global symptom checker applications in the context of common primary care presentations in Kenya, finding that none of the five platforms evaluated provided clinically appropriate recommendations for malaria — the leading cause of outpatient attendance in Kenyan public health facilities. This finding was of direct relevance to the SHES project, as it documented empirically the clinical inadequacy of existing platforms for Kenya's most prevalent primary care diagnosis and validated the necessity of a locally grounded knowledge base.

2.7 Research Gap

The literature review conducted in this chapter identified a consistent and substantive research and implementation gap at the intersection of health informatics, artificial intelligence, and Kenya's primary healthcare system. While the international literature on clinical decision support systems was extensive and methodologically sophisticated, the body of published work addressing the design, development, and evaluation of AI-powered clinical decision support systems specifically contextualized for Kenya — drawing on Kenya's clinical guidelines, disease epidemiology, medicines list, and healthcare infrastructure — was extremely limited.

The gap could be characterized along four dimensions. First, there was a contextualization gap: existing platforms were designed for high-income healthcare contexts and lacked the clinical knowledge, disease coverage, and medication grounding necessary for reliable operation within Kenya's healthcare system. Second, there was a comprehensiveness gap: no existing platform provided an integrated suite of triage, chronic disease monitoring, mental health support, medication management, drug interaction checking, and laboratory results interpretation within a single, user-accessible platform. Third, there was a transparency gap: commercially available probabilistic AI systems offered limited insight into their reasoning processes, limiting the clinical accountability and user trust necessary for adoption in settings where health technology was novel and trust was still being established. Fourth, there was a

compliance gap: no existing platform was designed in explicit alignment with the Kenya Data Protection Act, 2019, the Kenya National Clinical Guidelines, or the Kenya Essential Medicines List.

SHES was developed in direct response to this four-dimensional gap. By designing an AI health platform grounded in Kenya's own clinical standards, integrating a comprehensive suite of health management modules within a single platform, employing a transparent and auditable rule-based inference engine, and implementing a security architecture explicitly compliant with Kenya's data protection legislation, SHES made an original and practically significant contribution to the Kenyan health informatics landscape.

2.8 Chapter Summary

This chapter presented a comprehensive review of the literature pertaining to health informatics, clinical decision support systems, digital health in sub-Saharan Africa, and related commercial health technology platforms. The review traced the theoretical foundations of expert systems from MYCIN through modern CDSSs, contextualized the SHES project within Kenya's documented healthcare access challenges, and conducted a critical comparative analysis of Babylon Health, Ada Health, and WebMD Symptom Checker against SHES across twelve functional and contextual dimensions. The analysis identified a substantive research and implementation gap characterized by inadequate clinical contextualization, limited functional comprehensiveness, insufficient reasoning transparency, and absence of Kenya-specific regulatory compliance in existing platforms. This gap provided the empirical and theoretical justification for the development of SHES as an original contribution to the Kenyan health informatics landscape. Chapter Three proceeded to describe the methodology employed in the design and development of SHES.

END OF CHAPTER TWO

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter presented a detailed account of the research and development methodology employed in the design, development, and evaluation of the Smart Health Expert System (SHES). It described the philosophical orientation of the research, the software development methodology adopted, the procedures through which system requirements were elicited and documented, the data collection methods employed, and the analytical frameworks applied to transform raw research data into actionable system design specifications. The chapter further presented the system requirements — both functional and non-functional — derived from the requirements elicitation process, establishing the evaluative framework against which the implemented system was subsequently assessed in Chapter Six.

3.2 Research Design and Philosophical Orientation

The research adopted a pragmatic philosophical orientation, which Creswell and Creswell (2018) described as a worldview that focused on the practical consequences of ideas and actions rather than adherence to a single paradigm of truth or method. Pragmatic orientation was particularly appropriate for applied computing research — such as health informatics system development — where the primary criterion for methodological choice was fitness for purpose rather than philosophical purity.

Within this pragmatic orientation, the research employed a mixed-methods design that combined quantitative data collection through structured questionnaires with qualitative inquiry through semi-structured interviews and observational fieldwork. This methodological triangulation was adopted because the research questions addressed in the project spanned both quantifiable dimensions — such as the frequency of specific healthcare access challenges and the prevalence of digital technology ownership among target users — and qualitative dimensions, including the nature of clinical workflow pain points, the experiential barriers to healthcare access, and the preferences and expectations of healthcare professionals regarding digital health tools.

The research further employed an applied research design in which the primary output was a functioning software artefact — SHES — rather than a theoretical proposition or a generalized empirical finding. Applied research in software engineering, as characterized by Wohlin et al. (2012), sought to develop solutions to identified practical problems through the systematic application of established technical knowledge, and its quality was assessed through the functional correctness, technical robustness, and practical utility of the artefact produced. This framing positioned the SHES project firmly within the tradition of design science research in information systems, as articulated by Hevner et al. (2004), which required the construction of a novel, purposeful artefact addressing an identified organizational or social problem and the rigorous evaluation of that artefact against defined performance criteria.

3.3 Software Development Methodology: Agile

3.3.1 Overview of the Agile Methodology

The SHES project was developed using the Agile Software Development Methodology, guided by the principles articulated in the Agile Manifesto (Beck et al., 2001). Agile methodology prioritized individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan. These principles were operationalized through an iterative, sprint-based development cycle in which the system was built incrementally, with each sprint producing a tested, demonstrable increment of working software.

Specifically, the project adopted practices aligned with the Scrum framework — a widely used Agile implementation framework described by Schwaber and Sutherland (2020) — including time-boxed sprints of two to three weeks, a prioritized product backlog maintained throughout the project, sprint planning sessions at the

commencement of each sprint, and sprint retrospectives at the conclusion of each sprint to review progress, identify impediments, and adjust the development plan for subsequent iterations. Given that the project was developed by a single researcher with supervisory oversight rather than a cross-functional development team, certain Scrum ceremonies were adapted to the single-developer context while preserving the core principles of iterative delivery and continuous reflection.

3.3.2 Justification for Agile Methodology

The selection of Agile methodologies over alternative approaches — particularly the Waterfall model and the Spiral model — was justified based on four considerations specific to the SHES project context.

First, the nature of the problem domain demanded iterative requirements for refinement. Health informatics systems were characterized by complex, multistakeholder requirements that evolved as domain knowledge deepened through research and user engagement. The Agile approach accommodated this evolutionary character by treating requirements as a living backlog that was continuously refined, reprioritized, and extended throughout the development process — in contrast to the Waterfall model's assumption that requirements could be fully and definitively specified at project inception.

Second, the technical complexity of the SHES system — spanning a forward-chaining inference engine, a multi-module REST API, a React TypeScript frontend, field-level encryption, and a comprehensive test suite — demanded a development approach that facilitated early and continuous integration testing. The Agile practice of delivering working software at the end of each sprint enabled integration issues to be identified and resolved incrementally rather than accumulating into a high-risk final integration phase, as was characteristic of Waterfall development.

Third, the alignment between Agile's emphasis on user collaboration and the clinical validation requirements of a health decision support system was a compelling consideration. Clinical decision support systems that failed to incorporate iterative clinician and user feedback in their design processes risked producing technically correct but clinically inappropriate outputs — a failure mode documented in the health informatics literature (Ash et al., 2004). Agile's facilitation of regular user feedback loops provided a structural mechanism for incorporating clinical domain expertise throughout the development process rather than only at defined review milestones.

Fourth, the Agile methodology's inherent support for scope flexibility was particularly valuable in the context of an undergraduate research project operating within a fixed time constraint. As development progressed and the technical complexity of certain modules became clearer, the prioritized backlog framework enabled scope adjustments — deferring lower-priority features to future development cycles while ensuring that core, high-priority functionality was delivered to a high standard within the available development time.

3.3.3 Sprint Plan and Development Schedule

The SHES project was executed across nine sprints spanning twenty-two weeks. Table 3.1 presented the sprint plan, including the objectives, deliverables, and timeline for each sprint.

Table 3.1: SHES Agile Sprint Plan

Sprint	Duration	Objectives	Deliverables
1	Weeks 1–2	Requirements elicitation, stakeholder interviews, problem definition	Project proposal, system requirements specification draft
2	Weeks 3–4	System architecture design, database schema, UI wireframes	Architecture diagram, ERD, wireframe prototypes
3	Weeks 5–7	Backend setup: Django project, authentication module, JWT, database migrations	Working authentication API, PostgreSQL schema, user registration and login endpoints
4	Weeks 8–10	Inference engine development: knowledge base encoding, forward-chaining logic, red flag detection	Functional triage API with 14-condition KB and 12 red flag patterns; unit tests
5	Weeks 11–13	Medication management, drug interaction checker, chronic disease tracking modules	KEML catalogue API, interaction checker, glucose and BP logging endpoints with auto-classification
6	Weeks 14–15	Mental health module, lab results interpreter, patient profile module	Mood tracking API, coping strategy engine, lab interpreter, patient profile CRUD endpoints
7	Weeks 16–18	Frontend development: React SPA, all pages, component library, TanStack Query integration	Fully functional React frontend with all 9 pages, responsive design, JWT interceptors

8	Weeks 19–20	System integration, end-to-end testing, UAT, bug fixing, security hardening	Integrated system, 165+ test cases passing, UAT sign-off, security audit report
9	Weeks 21–22	Documentation, Docker deployment configuration, final report writing	Complete project report, Docker Compose deployment stack, README, API reference

Source: Researcher's own design (2026)

The sprint plan reflected a deliberate sequencing strategy in which foundational infrastructure components — authentication, database schema, and project scaffolding — were established in early sprints to provide a stable development platform for the more complex functional modules developed in subsequent sprints. The inference engine, as the most clinically critical and technically novel component of the system, was allocated a dedicated sprint to ensure sufficient time for knowledge base design, inference logic implementation, and comprehensive unit testing.

3.4 Requirements Gathering Procedures

Requirements gathering for the SHES project was conducted through a systematic, multi-method process designed to capture the diverse perspectives of the system's primary stakeholder groups: patients and general users of healthcare services, clinical professionals engaged in primary care, and health system administrators. The requirements gathering process was conducted during Sprints 1 and 2 and was structured to address three categories of requirements information: problem domain knowledge, user needs and expectations, and technical constraints.

3.4.1 Stakeholder Identification

Stakeholder analysis was conducted at the commencement of the requirements gathering phase to identify all parties with a material interest in the SHES system, the nature of their interest, and the appropriate method for capturing their requirements. Five primary stakeholder groups were identified:

- Patients: adult individuals seeking preliminary health guidance, symptom assessment, or chronic disease self-monitoring support. Their primary requirements are related to ease of use, accuracy of clinical outputs, privacy of health data, and accessibility on commonly available devices.
- Clinical Officers and Nurses: frontline healthcare professionals who constituted potential users of the system's doctor-facing dashboard and whose domain

expertise was essential to the clinical validation of the inference engine's outputs and recommendations.

- Healthcare Administrators: managers of health facilities who were concerned with the system's potential to reduce non-urgent outpatient attendance and its compliance with health data regulations.
- Preventive Healthcare Seekers: health-conscious individuals who sought tools for proactive health monitoring, chronic disease self-management, and medication adherence support.
- The Kenya Ministry of Health (indirect stakeholder): whose published clinical guidelines, essential medicines list, and data protection legislation constituted normative requirements that the system was obligated to satisfy.

3.4.2 Data Collection Methods

Four complementary data collection methods were employed to gather requirements from the identified stakeholder groups. Table 3.2 summarizes each method, its target participants, the nature of data sought, and the instrument employed.

Table 3.2: Data Collection Methods and Instruments

Method	Target Participants	Data Sought	Instrument / Mode
Structured Interviews	3 registered nurses, 2 clinical officers, 1 hospital IT administrator	Clinical workflow pain points, triage process, NCD monitoring gaps, patient non-compliance patterns	Semi-structured interview guide; face-to-face and video call
Online Questionnaire	45 adult respondents (patients and health-aware public)	Healthcare access challenges, digital health willingness, symptom-checking behavior, medication adherence	Google Forms; 22-item Likert-scale and open-ended questionnaire
Observation	Outpatient department of a Level 4 public health facility	Patient flow patterns, triage procedures, queue dynamics, clinician workload	Non-participant structured observation; field notes
Document / Literature Review	Published guidelines, government reports, peer-reviewed journals	Kenya clinical guidelines, disease prevalence data, KEMIL, DPA 2019, health informatics best practice	Google Scholar, PubMed, MOH Kenya publications, WHO technical documents

Source: Researcher's own design (2026)

3.4.3 Interviews

Semi-structured interviews were conducted with six clinical and administrative informants selected purposively based on their direct professional engagement with the primary care challenges that SHES was designed to address. The interview guide comprised three thematic sections: an exploration of current clinical workflow and triage practices; an assessment of the specific challenges arising from delayed diagnosis, NCD management, and medication adherence; and a forward-looking inquiry into the features, design principles, and data requirements that informants considered most important in a digital clinical decision support tool.

Each interview lasted between forty-five and sixty minutes. Interviews were conducted face-to-face at the informant's place of work where possible and via video call where logistical constraints made in-person meetings impracticable. All interviews were conducted with the informed consent of the participants, who were assured of the confidentiality of their individual responses and the anonymisation of any identifying

information in the final report. Interviews were recorded with participants' permission and subsequently transcribed for thematic analysis.

Thematic analysis of the interview transcripts was conducted using the six-phase framework described by Braun and Clarke (2006): familiarization with the data through repeated reading; generation of initial codes; searching for themes; reviewing themes; defining and naming themes; and producing the final analysis. Three primary themes emerged from the analysis: the critical inadequacy of existing triage mechanisms for managing patient flow in primary care settings; the absence of accessible, contextually appropriate tools for chronic disease self-monitoring; and the significant informational barrier between patients and the clinical significance of their own health data — particularly laboratory results and vital sign readings.

3.4.4 Online Questionnaire

A structured online questionnaire was administered to forty-five respondents recruited through purposive sampling from the researcher's academic network, healthcare contacts, and professional associations. The questionnaire was developed on the Google Forms platform and comprised twenty-two items distributed across four sections: demographic information; current health-seeking behavior and access challenges; digital technology ownership, literacy, and health application usage; and willingness-to-use and feature preference ratings for a proposed digital health platform.

Items in the third and fourth sections employed five-point Likert scales (1 = Strongly Disagree to 5 = Strongly Agree; 1 = Extremely Unlikely to 5 = Extremely Likely), with open-ended follow-up items enabling respondents to elaborate on their scaled responses. Descriptive statistical analysis of the Likert-scale items was conducted using measures of central tendency and frequency distribution. Thematic analysis was applied to the qualitative open-ended responses. Key findings from the questionnaire included: 78% of respondents reported having sought health information online prior to seeking formal clinical care; 71% expressed uncertainty about when symptoms warranted a formal clinical consultation; 84% expressed interest in a platform offering AI-based preliminary symptom assessment; and 67% reported having missed or forgotten at least one medication dose in the preceding month — a finding that directly informed the prioritization of the Medication Management module.

3.4.5 Observation

Non-participant structured observation was conducted at the outpatient department of a Level 4 public health facility over two half-day sessions. The observation protocol focused on four dimensions of clinical service delivery: patient arrival and initial triage processes; queue management and waiting time patterns; the nature and distribution of presenting complaints; and clinician-patient interaction patterns, with particular attention to the proportion of consultations involving symptom inquiries that could potentially have been addressed through a structured self-care tool.

Field notes recorded during the observation sessions were subsequently analysed using a descriptive-interpretive approach, with particular attention to the identification of operational patterns consistent with the structural problems identified in the interview and questionnaire data. Observations confirmed the clinical informants' accounts of the predominance of non-urgent presentations in the outpatient queue, the limited time available for individual clinician-patient consultation given throughput pressures, and the systematic absence of structured triage instruments capable of stratifying patients by urgency prior to clinical consultation.

3.4.6 Literature Review as a Data Source

As documented in Chapter Two, the systematic literature review conducted during the requirements gathering phase served a dual function: it provided theoretical and empirical grounding for the problem definition and solution design, and it constituted a primary source of normative requirements derived from authoritative clinical guidelines and regulatory frameworks. Specifically, the Kenya National Clinical Guidelines (Ministry of Health Kenya, 2016), the Kenya Essential Medicines List (Ministry of Health Kenya, 2022), the WHO Primary Care Protocols (WHO, 2020), the JNC-8 blood pressure classification guidelines (James et al., 2014), and the Kenya Data Protection Act, 2019, were each treated as requirement sources that established mandatory parameters for the clinical content, data security, and operational design of SHES.

3.5 System Requirements Specification

The requirements gathered through the methods described in Section 3.4 were synthesized and formalized into a System Requirements Specification (SRS) that served as the foundational document for the system design and implementation phases. Requirements were categorized as functional requirements — defining what the system should do — and non-functional requirements — defining the performance, security, usability, and maintainability standards the system should meet.

Requirements were further classified by priority using the MoSCoW prioritization framework (Clegg & Barker, 1994): Must Have (essential, non-negotiable), Should Have (important but not critical to core function), Could Have (desirable if time permitted), and Won't Have (explicitly excluded from the current version). This classification informed sprint backlog prioritization throughout the Agile development cycle.

3.5.1 Functional Requirements

Table 3.3 presented the functional requirements of the SHES system as derived from the requirements elicitation process.

Table 3.3: SHES Functional Requirements

FR No.	Functional Requirement	Priority
FR-01	The system shall allow users to register with an email address, password, and role selection (Patient, Doctor, Admin).	Must Have
FR-02	The system shall authenticate users using JSON Web Tokens (JWT) with access token refresh and logout blacklisting.	Must Have
FR-03	The system shall allow patients to submit symptoms with name, severity (1–10), duration in days, and optional body location.	Must Have
FR-04	The system shall scan submitted symptoms against 12 emergency red flag patterns and route Emergency-level cases immediately.	Must Have
FR-05	The system shall match submitted symptoms against a 14-condition clinical knowledge base using a forward-chaining inference engine.	Must Have
FR-06	The system shall classify triage output into one of four urgency levels: Emergency, Doctor Visit, Self-Care, or Undetermined.	Must Have
FR-07	The system shall provide plain-language triage recommendations, matched condition names, match ratios, and personalized home-care tips.	Must Have
FR-08	The system shall allow patients to log blood glucose readings with context (fasting, post-meal, random, bedtime) and auto-classify against WHO targets.	Must Have
FR-09	The system shall allow patients to log blood pressure readings and auto-classify against JNC-8 categories.	Must Have
FR-10	The system shall display 14-reading longitudinal trend charts for blood glucose and blood pressure.	Must Have
FR-11	The system shall allow patients to search the Kenya Essential Medicines List catalogue and manage their personal medication list.	Must Have
FR-12	The system shall perform pairwise drug interaction checks across selected medications with severity ratings.	Must Have
FR-13	The system shall allow patients to log daily mood entries with score, emotion tags, and optional journal notes.	Should Have
FR-14	The system shall recommend evidence-based coping strategies following each mood log entry.	Should Have
FR-15	The system shall allow patients to submit laboratory test results and receive plain-language interpretation against MOH reference ranges.	Must Have

FR-16	The system shall maintain a patient's medical profile including blood group, allergies, chronic conditions, and emergency contact.	Must Have
FR-17	The system shall maintain a complete triage session history accessible by the authenticated patient.	Should Have
FR-18	The system shall provide a 7-day summary dashboard displaying average vitals, latest triage results, and mood trends.	Should Have

Source: Researcher's own compilation based on stakeholder requirements elicitation (2026)

3.5.2 Non-Functional Requirements

Table 3.4 presented the non-functional requirements of the SHES system, specifying the quality attributes that the implemented system was required to satisfy.

Table 3.4: SHES Non-Functional Requirements

NFR No.	Category	Requirement
NFR-01	Performance	API response time for triage assessment shall not exceed 2 seconds under normal load conditions.
NFR-02	Performance	The frontend SPA shall achieve a Lighthouse performance score of not less than 80 on desktop browsers.
NFR-03	Security	All patient health data shall be transmitted over HTTPS using TLS encryption in the production environment.
NFR-04	Security	Sensitive health data fields shall be encrypted at rest using AES-128-CBC (Fernet) field-level encryption.
NFR-05	Security	The system shall implement rate limiting: 10 auth requests/hour, 30 triage requests/hour, 200 general API requests/hour.

NFR-06	Security	The system shall comply with the Kenya Data Protection Act, 2019, with audit logging of all API transactions.
NFR-07	Usability	The user interface shall be responsive and fully functional on viewport widths of 375px and above.
NFR-08	Usability	Triage outputs shall be written in plain English at a reading level accessible to adult users with a secondary education.
NFR-09	Reliability	The system shall achieve an uptime of not less than 99% during the evaluation period.
NFR-10	Maintainability	Clinical knowledge base content shall be modifiable through JSON file editing without code changes.
NFR-11	Scalability	The system architecture shall support horizontal scaling through containerisation with Docker and Docker Compose.
NFR-12	Interoperability	The backend shall expose a RESTful API conforming to HTTP/JSON standards, enabling future integration with external systems.

Source: Researcher's own compilation based on stakeholder requirements elicitation and regulatory review (2026)

3.6 Design Procedures

Following the completion of the requirements gathering and specification phase, the design process was initiated in Sprint 2. The design phase addressed four interconnected levels of system design: system architecture, database schema, module design, and user interface design. Each design level was informed by the requirements specification produced in the preceding phase and was subject to iterative refinement as implementation revealed new constraints or opportunities not anticipated during the initial design.

3.6.1 System Architecture Design

The system architecture was designed as a three-tier client-server architecture comprising a React TypeScript single-page application (SPA) frontend, a Django REST Framework API backend, and a PostgreSQL relational database. This architectural pattern was selected on the basis of its alignment with the system's scalability, maintainability, and deployment requirements. The separation of concerns between the presentation layer (React frontend), business logic layer (Django backend and inference engine), and data persistence layer (PostgreSQL) facilitated independent development

and testing of each tier and supported horizontal scaling of the backend under increased load.

The backend API was designed as a RESTful service conforming to the Representational State Transfer architectural constraints articulated by Fielding (2000): statelessness, uniform interface, resource-based addressing, and layered system structure. All API endpoints were versioned under the `/api/v1/` prefix to support future API evolution without breaking existing clients. JWT-based stateless authentication was adopted to enable horizontal backend scaling without the session state synchronization complexity associated with cookie-based authentication.

The inference engine was designed as a standalone Python class — `InferenceEngine` — encapsulated within the triage application module, operating independently of the web framework to facilitate unit testing and potential future extraction as a microservice. The engine's knowledge base was implemented as three JSON files — `red_flags.json`, `conditions.json`, and `recommendations.json` — loaded at application startup and cached in memory, enabling clinical content updates through file editing without application restart.

The system architecture diagram below illustrates the high-level component relationships and request lifecycle of the SHES system.

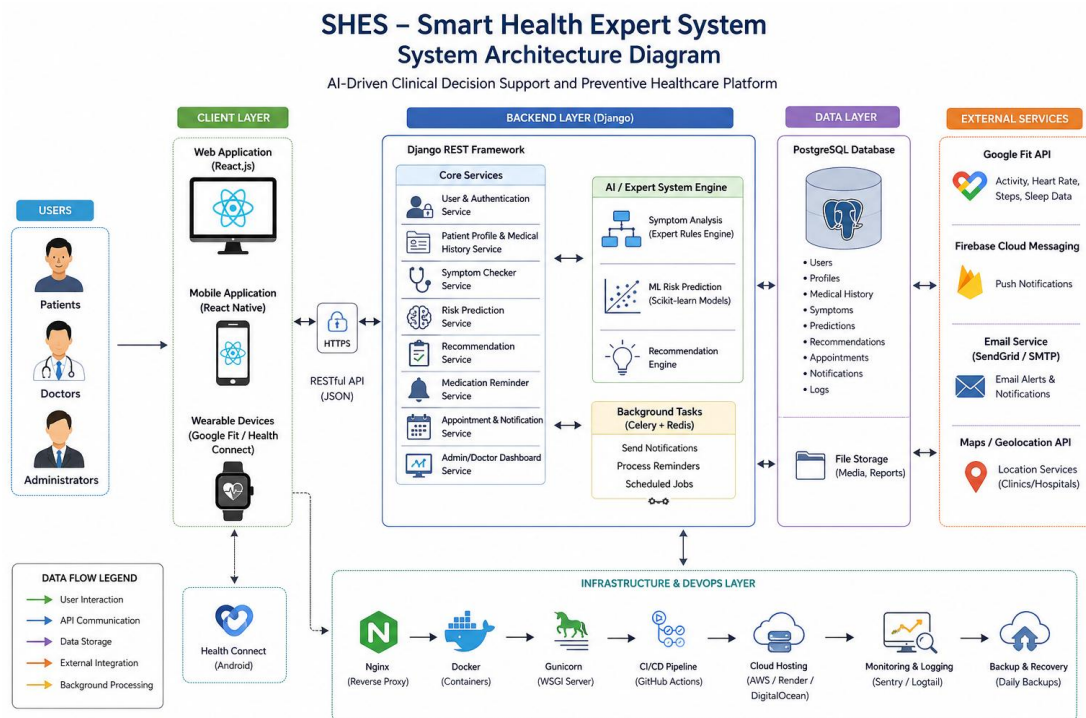


Figure 3.1: SHES System Architecture Diagram

3.6.2 Database Design

The database schema was designed using a relational model implemented in PostgreSQL 15. The schema comprised eleven primary tables, each representing a

distinct entity within the system's domain model: authentication_user (the custom User model extending Django's AbstractBaseUser), authentication_patientprofile, triage_triagesession, triage_symptom, medications_medication, medications_druginteraction, medications_patientmedication, chronic_tracking_glucosereading, chronic_tracking_bloodpressurereading, mental_health_moodentry, mental_health_copingstrategy, and lab_results_labresult.

UUID primary keys were employed for all patient-facing data tables to prevent sequential ID enumeration attacks — a security practice recommended by the OWASP API Security Top 10 (OWASP, 2023). Foreign key relationships enforced referential integrity between related entities, with cascading deletion rules ensuring that the removal of a patient account triggered the appropriate cleanup of associated health records. Field-level Fernet encryption was applied to designated sensitive fields — including allergy records, chronic condition narratives, and journal notes — prior to database persistence.

The Entity Relationship Diagram below depicts the relationships between the primary tables of the SHES database schema.

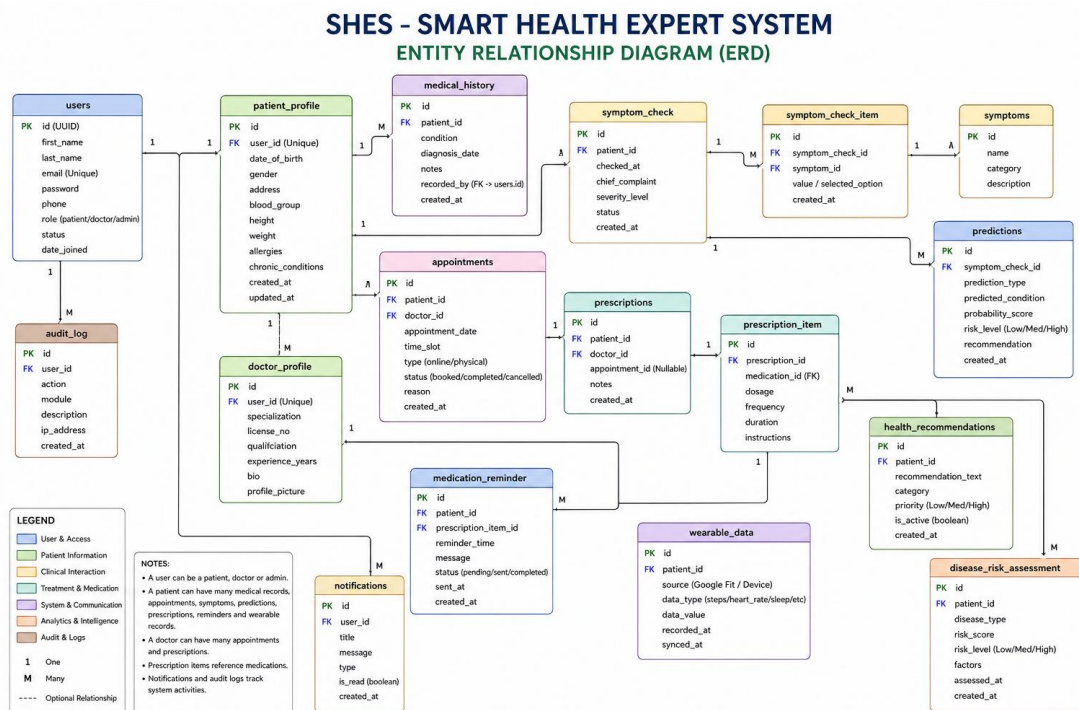


Figure 3.2: SHES Entity Relationship Diagram

3.6.3 User Interface Design

User interface design was conducted in accordance with the health literacy and human-computer interaction principles reviewed in Chapter Two. Wireframes were developed for each of the nine primary pages of the application — Login, Register, Dashboard, Symptom Triage, Triage History, Medications, Chronic Tracking, Mental Health, Lab

Results, and Profile — and were reviewed against the system requirements specification and the usability dimensions identified by Cengiz and Ozbay (2022).

The visual design employed Tailwind CSS's utility-first styling framework to implement a consistent, accessible design system across all application pages. A sidebar navigation layout was selected to provide persistent access to all major modules without requiring page reloads, consistent with the single-page application pattern. Color coding was applied systematically to urgency levels — red for Emergency, amber for Doctor Visit, green for Self-Care, and grey for Undetermined — to communicate clinical significance through visual redundancy that supported users with varying levels of health and digital literacy.

The UI wireframes and prototype screenshots are presented in Chapter Five (System Design) and Chapter Six (Implementation).

3.7 Data Analysis Methods

The analysis of data collected through the requirements gathering instruments proceeded through two parallel analytical tracks corresponding to the quantitative and qualitative nature of the data collected.

Quantitative data from the online questionnaire — specifically the Likert-scale items measuring healthcare access challenges, digital technology ownership, and feature preference ratings — was analyzed using descriptive statistics. Frequency distributions and percentage calculations were applied to ordinal Likert-scale responses to identify patterns of consensus and divergence among respondents. Mean scores and standard deviations were calculated for multi-item constructs to assess the strength and consistency of respondents' attitudes toward the proposed system features. These statistical summaries were used to prioritise requirements in the product backlog, with features rated most highly by respondents elevated to Must Have priority and those attracting lower or more variable ratings classified as Should Have or Could Have.

Qualitative data from interviews and open-ended questionnaire responses was analyzed using thematic analysis as described by Braun and Clarke (2006). The six-phase process of familiarization, initial coding, theme identification, theme review, theme definition, and write-up was applied to produce a structured thematic framework that organized the qualitative findings into coherent categories of user need, operational challenge, and system expectation. The thematic framework was used to inform the narrative framing of the problem statement and the design rationale for specific system features, particularly the clinical content of the inference engine, the plain-language design of triage outputs, and the selection of the fourteen conditions covered by the knowledge base.

Data from the observational sessions was analyzed using a descriptive-interpretive approach in which field notes were reviewed against the a priori thematic framework derived from the literature review and elaborated through emerging patterns observed

in the field. The observational data served primarily as a triangulation source, confirming or challenging findings from the interview and questionnaire instruments and providing contextual grounding for the quantitative estimates of patient flow patterns and non-urgent consultation proportions.

The integration of findings from the three analytical tracks was achieved through a process of triangulation in which convergent findings across methods were treated as providing strong evidential support for the corresponding system requirement, while divergent findings prompted additional inquiry and discussion with clinical informants to resolve the discrepancy before the requirement was finalized.

3.8 Ethical Considerations

The research was conducted in adherence to established principles of research ethics applicable to applied computing research involving human participants. Informed consent was obtained from all interview and questionnaire participants prior to data collection, with participants provided with a clear explanation of the research purpose, the voluntary nature of their participation, and their right to withdraw at any time without consequence. Participant anonymity was maintained throughout the data collection, analysis, and reporting process, with no identifying information included in the final report.

The clinical nature of the SHES system raised specific ethical considerations pertaining to the potential for harm arising from inaccurate triage outputs. These considerations were addressed through several design-level safeguards: the system's consistent framing of all triage outputs as preliminary and advisory rather than definitive diagnostic conclusions; the inclusion of explicit instructions to seek professional clinical care for all presentations classified at Emergency or Doctor Visit urgency levels; the integration of Kenya-specific emergency contact numbers (999 and Kenya Red Cross 0800 723 253) in Emergency-level outputs; and the implementation of a disclaimer visible at all triage output screens communicating that SHES was a clinical decision support tool and not a replacement for professional medical advice.

Data privacy and security ethical obligations were addressed through the technical security architecture described in the system requirements specification: field-level encryption of sensitive health data, JWT-based authentication with token blacklisting on logout, audit logging of all API transactions, and explicit design compliance with the Kenya Data Protection Act, 2019. No patient health data was collected from real patients during the development or testing phases; all testing was conducted using synthetic data generated by the researcher and automated test data factories.

3.9 Chapter Summary

This chapter presented a comprehensive account of the methodology employed in the SHES project. It described the pragmatic, mixed-methods research design orientation; provided a detailed justification for the adoption of the Agile Software Development Methodology over alternative approaches; documented the nine-sprint development schedule; described the four data collection methods employed in requirements gathering — interviews, questionnaire, observation, and literature review; presented the formal system requirements specification comprising eighteen functional requirements and twelve non-functional requirements; outlined the design procedures applied to system architecture, database schema, and user interface design; and described the quantitative and qualitative analytical methods applied to requirements data. Ethical considerations pertaining to clinical safety, participant rights, and data privacy were also addressed. Chapter Four proceeded to present a detailed analysis of the existing system context in Kenya and the specific system analysis artefacts — use cases, context diagrams, and data flow diagrams — derived from this requirements analysis.

END OF CHAPTER THREE

CHAPTER FOUR: SYSTEM ANALYSIS

4.1 Introduction

This chapter presented a comprehensive analysis of the existing healthcare system context in Kenya and derived from that analysis the formal system specifications and modelling artefacts that guided the design and implementation of SHES. The chapter examined the structural weaknesses of Kenya's current primary healthcare consultation system from a systems analysis perspective, documented the functional and non-functional requirements in applied form, presented the use case analysis defining system behavior from the user's perspective, and provided academic explanations of the system's context diagram and data flow diagrams. The analysis artefacts produced in this chapter constituted the analytical bridge between the problem domain explored in Chapters One and Two and the solution domain detailed in Chapters Five and Six.

4.2 Analysis of Existing Healthcare Consultation Challenges in Kenya

4.2.1 The Current System: Manual, Facility-Centered Primary Care

The existing system of primary healthcare consultation in Kenya was characterized by a predominantly manual, facility-centered model that concentrated clinical decision-making at the point of physical encounter between a patient and a healthcare professional. This model, while appropriate in the context of complex or high-acuity presentations, was applied indiscriminately to the full spectrum of primary care demand — including large volumes of low-acuity, self-limiting conditions that could reasonably have been managed through structured self-care guidance. The consequences of this indiscriminate facility-dependence were measurable and severe: chronic overcrowding of outpatient departments, extended waiting times that averaged between two and four hours in Level 4 and 5 public facilities (Kenya Health Service Delivery Report, 2022), clinician consultation times compressed to an average of under four minutes per patient due to throughput pressure, and a systemic inability to prioritize genuinely urgent presentations above the noise of non-urgent volume.

The structural weaknesses of the current system were analyzed across seven operational dimensions. Table 4.1 presented this analysis, documenting the current state, its operational impact, and the specific SHES response for each dimension.

Table 4.1: Current System Weaknesses and SHES Responses

Weakness Area	Current State	Impact	SHES Response
Triage Mechanism	Manual, verbal triage by nurses without standardized tools	Inconsistent urgency classification; non-urgent cases delay emergency care	Structured 4-level automated urgency triage with 12 red flag patterns

Symptom Information Access	Patients rely on informal networks or uncontextualized global websites	Delayed care-seeking; inappropriate self-medication; unnecessary hospital visits	Localized AI inference engine with 14 Kenya-relevant conditions
Chronic Disease Monitoring	BP and glucose readings taken episodically during facility visits only	Undetected deterioration between visits; reactive rather than preventive management	Continuous self-monitoring with WHO/JNC-8 auto-classification and trend charts
Medication Management	Paper prescription cards; no digital reminder or interaction checking system	Missed doses; undetected dangerous drug combinations in polypharmacy patients	Digital medication list with interaction checker grounded in KEML
Mental Health Support	No digital mental health tool in Kenya's primary care digital ecosystem	85%+ treatment gap; mental illness undetected until crisis presentation	Integrated mood tracker with coping strategies and 14-day trend analysis
Lab Results Interpretation	Patients receive printed results with no accessible explanation	Patient confusion; unnecessary anxiety; inability to act on results	Plain-language interpreter for 9 common tests against MOH reference ranges
Data Privacy	Paper records with minimal access controls in many facilities	Patient records accessible to unauthorized staff; no audit trail	Fernet field-level encryption; JWT auth; full audit logging per DPA 2019

Source: Researcher's own analysis based on fieldwork observation and stakeholder interviews (2026)

4.2.2 The Digital Health Gap in Kenya's Primary Care Ecosystem

Beyond the operational weaknesses of the physical care system, the analysis identified a critical gap in Kenya's digital health landscape: the absence of an integrated, locally grounded digital primary care platform capable of providing preliminary clinical guidance, chronic disease self-monitoring, medication management, and mental health support through a single accessible interface. As established in the literature review, the available global platforms — Babylon Health, Ada Health, and WebMD — were calibrated for non-Kenyan clinical contexts and did not address the specific clinical, linguistic, regulatory, and economic characteristics of Kenya's healthcare environment. The KEMR electronic health record system deployed in public facilities, while valuable for clinical documentation, was a provider-facing tool with no patient-facing clinical decision support capability.

The digital health gap was further characterized by the fragmentation of the limited existing digital health tools: medication reminder applications, general wellness apps, and health information websites existed as separate, unintegrated tools with no shared data infrastructure or coordinated clinical logic. A patient managing hypertension and diabetes simultaneously, seeking to track their vitals, understand a laboratory result, check a drug interaction, and receive a mood-based coping strategy, required a minimum of four separate applications — none of which was grounded in Kenya's clinical guidelines. SHES was designed to collapse this fragmented landscape into a single, integrated, clinically coherent platform.

4.3 Use Case Analysis

Use case analysis was conducted to define the system's behavior from the perspective of its users, capturing the functional interactions between the system's three actor categories — Patient, Doctor/Administrator, and System (automated processes) — and the system's functional modules. The use case analysis was grounded in the requirements specification documented in Chapter Three and served as the primary input to the system design activities documented in Chapter Five.

4.3.1 System Actors

Three primary actors were identified in the SHES system:

- **Patient:** An authenticated adult user who accessed the system to perform symptom triage, log vital signs, manage medications, track mood, submit laboratory results, and maintain their personal health profile. The Patient actor was the primary user of all health management modules.
- **Doctor / Healthcare Administrator:** An authenticated clinical or administrative user who accessed the system's management dashboard to review patient records, monitor aggregate health trends, and oversee system usage. This actor had read-only access to patient health data and administrative control over user account management.
- **System (Automated Actor):** The backend processing components of SHES — including the inference engine, the drug interaction checker, the lab results interpreter, and the chronic disease auto-classifier — that were triggered by patient actions and executed without direct human intervention to produce structured clinical outputs.

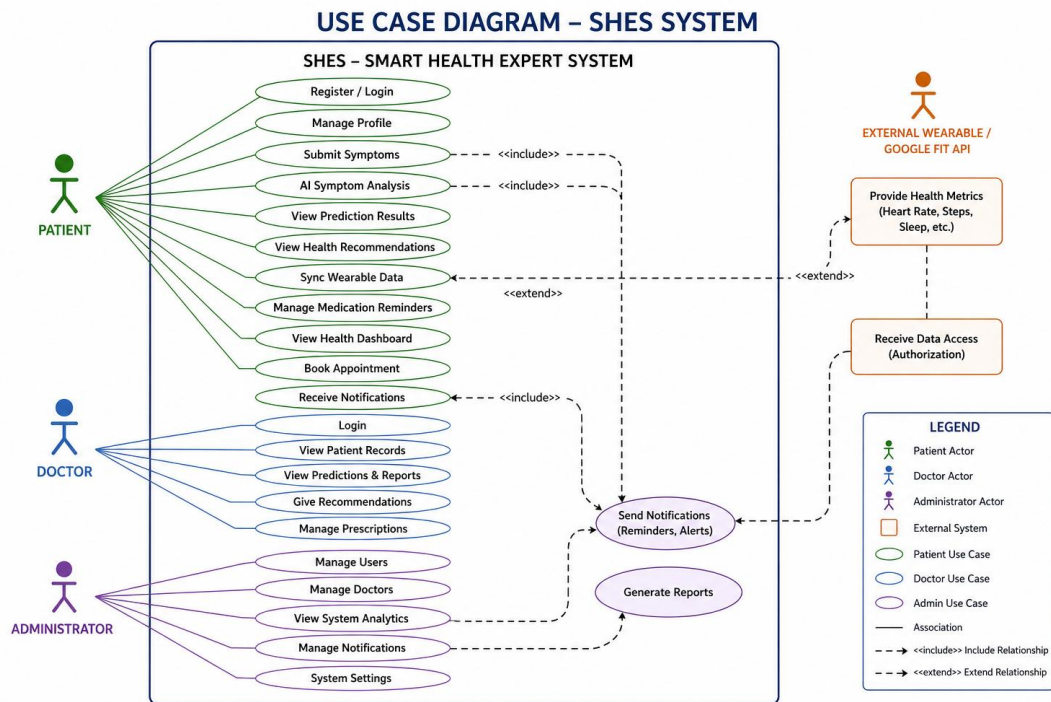


Figure 4.1: SHES System Use Case Diagram

Figure 4.1 depicted the complete set of use cases supported by the SHES system, showing the association between each actor and the use cases they initiated or participated in. The diagram illustrated the breadth of the system's functional scope and the clear separation of concerns between patient-facing health management functions and administrative oversight functions.

4.3.2 Detailed Use Case Descriptions

The following use case descriptions provided the detailed specifications for the six most clinically significant use cases in the SHES system.

Use Case 1: Perform Symptom Triage Assessment

Field	Description
Use Case ID	UC-01
Use Case Name	Perform Symptom Triage Assessment
Actor(s)	Patient (Primary); System/Inference Engine (Secondary)
Preconditions	The patient is authenticated and has a confirmed account. Patient has one or more symptoms to report.

Trigger	Patient navigates to the Symptom Triage page and initiates a new assessment session.
Main Flow	1. Patient enters symptom name, severity rating (1–10), duration in days, and optional body location.2. Patient adds additional symptoms if applicable.3. Patient submits the symptom assessment form.4. System passes the submitted symptom list to the InferenceEngine.5. InferenceEngine scans for 12 emergency red flag patterns.6. InferenceEngine matches symptom profile against 14-condition knowledge base.7. InferenceEngine applies severity and duration escalation rules.8. System generates triage output: urgency level, clinical recommendation, matched conditions with ratios, and home-care tips.9. System persists the triage session to the database.10. System displays triage output to the patient.
Alternative Flow	If an emergency red flag detected at Step 5: System immediately routes to EMERGENCY urgency level and displays emergency contact numbers. Steps 6–7 are skipped.
Postconditions	A triage session record is stored in the database. The patient has received urgency classification and clinical guidance. If Emergency level, patient is directed to immediate care.
Exception Flow	If no symptoms match any condition and no red flags detected: System returns UNDETERMINED urgency level with advice to provide more symptoms or seek clinical consultation.

Use Case 2: Log and Monitor Chronic Disease Vitals

Field	Description
Use Case ID	UC-02
Use Case Name	Log and Monitor Chronic Disease Vitals
Actor(s)	Patient (Primary); System/Auto-Classifier (Secondary)
Preconditions	The patient is authenticated. Patient has blood glucose or blood pressure readings to log.
Trigger	Patient navigates to the Chronic Disease Tracking page.
Main Flow	1. Patient selects reading type (Blood Glucose or Blood Pressure).2. For glucose: Patient enters reading value (mmol/L) and measurement context (fasting, post-meal, random, bedtime).3. For BP: Patient enters systolic and diastolic values (mmHg) and pulse rate.4. System auto-classifies the reading against WHO glycaemic targets (glucose) or JNC-8 categories (BP).5. System persists the reading with timestamp and classification.6. System updates the longitudinal trend chart with the new reading.7. System displays classification label and trend chart showing up to 14 readings.
Alternative Flow	If reading value is outside physiologically plausible range: System displays a validation warning and requests re-entry.

Postconditions	Reading is stored in the database. Classification is displayed. Trend chart is updated.
Exception Flow	If fewer than 2 readings exist: Trend chart displays single data point with message indicating insufficient data for trend analysis.

Use Case 3: Check Drug Interactions

Field	Description
Use Case ID	UC-03
Use Case Name	Check Drug Interactions
Actor(s)	Patient (Primary); System/Interaction Checker (Secondary)
Preconditions	The patient is authenticated. Patient has at least two medications on their active medication list.
Trigger	Patient navigates to the Medications page and selects 'Check Interactions'.
Main Flow	1. System retrieves the patient's active medication list.2. System performs pairwise interaction checks across all medication combinations using the KEML-grounded interaction database.3. System generates a structured interaction report listing all detected interaction pairs, severity ratings (Mild/Moderate/Severe), and clinical guidance.4. System displays interaction report to patient with color-coded severity indicators.5. System recommends clinical consultation for any Severe-rated interaction.
Alternative Flow	If a patient has fewer than two medications: System displays a message indicating that at least two medications are required for interaction checking.
Postconditions	The patient has received a structured drug interaction report. Severe interactions are flagged for clinical follow-up.
Exception Flow	If no interactions are detected: System displays a confirmation message indicating no known interactions were identified in the KEML interaction database.

Use Case 4: Log Mood and Receive Coping Strategies

Field	Description
Use Case ID	UC-04
Use Case Name	Log Mood and Receive Coping Strategies
Actor(s)	Patient (Primary); System/Coping Strategy Engine (Secondary)

Preconditions	The patient is authenticated.
Trigger	Patient navigates to the Mental Health page and selects 'Log Today's Mood'.
Main Flow	1. Patient rates their mood on a 1–10 scale.2. Patient selects emotion tags from a predefined list (e.g., anxious, sad, stressed, content).3. Patient optionally enters a journal note.4. System saves the mood entry with timestamp.5. System selects evidence-based coping strategies matched to the patient's mood score and emotion tags.6. System displays coping strategies with explanations.7. System updates the 14-day mood trend chart.
Alternative Flow	If mood score is 1–3 (severe distress): System additionally displays crisis support resources including Befrienders Kenya contact details.
Postconditions	Mood entry is stored. Patient has received personalized coping strategy recommendations. The trend chart is updated.
Exception Flow	Not applicable.

Use Case 5: Submit and Interpret Laboratory Results

Field	Description
Use Case ID	UC-05
Use Case Name	Submit and Interpret Laboratory Results
Actor(s)	Patient (Primary); System/Lab Interpreter (Secondary)
Preconditions	The patient is authenticated. Patient has a laboratory result document or values to enter.
Trigger	Patient navigates to the Lab Results page and selects 'Add New Result'.
Main Flow	1. Patient selects test type from a list of 9 supported tests.2. Patient enters the test result value and measurement unit.3. System compares the value against MOH reference ranges for the selected test.4. System classifies the result as Normal, Below Normal, or Above Normal.5. System generates a plain-language interpretation explaining the result's significance.6. System provides actionable recommendations (e.g., dietary guidance, clinical follow-up).7. System stores the result with a timestamp.
Alternative Flow	If result value is critically abnormal: System flags the result as requiring urgent clinical review and displays a prominent clinical consultation recommendation.
Postconditions	The lab result is stored. Patient has received a plain-language interpretation. Results history is updated.
Exception Flow	Not applicable.

Use Case 6: Doctor Reviews Patient Dashboard

Field	Description
Use Case ID	UC-06
Use Case Name	Doctor Reviews Patient Dashboard
Actor(s)	Doctor / Administrator (Primary)
Preconditions	User is authenticated with Doctor or Administrator role.
Trigger	The doctor logs in and navigates to the Admin Dashboard.
Main Flow	1. System retrieves aggregate patient statistics: total registered patients, triage session counts by urgency level, average vitals across active patients.2. System displays a 7-day summary dashboard with key health indicators.3. The doctor selects a specific patient record to review.4. System displays the selected patient's triage history, vital sign logs, medication list, and mood trend.5. Doctor reviews data and may flag records requiring clinical follow-up.
Postconditions	The doctor has reviewed patient health data. Access events are logged in the audit trail.
Exception Flow	If patient records are not found System displays an appropriate error message.

4.4 Context Diagram

The context diagram depicted the SHES system at its highest level of abstraction — as a single processing entity surrounded by its external entities and the data flows between them. The context diagram served as the analytical boundary definition for the system, establishing which entities and data flows were within scope and which were external.

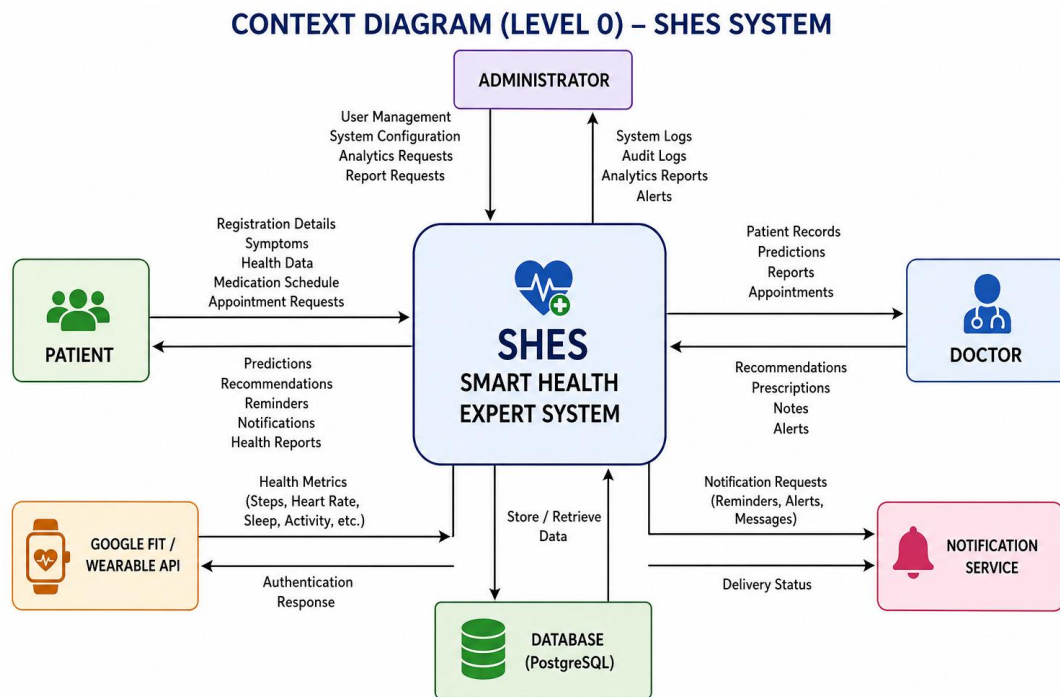


Figure 4.2: SHES System Context Diagram

As illustrated in Figure 4.2, the SHES system interacted with five external entities. The Patient entity submitted symptom data, vital sign readings, medication information, mood entries, and laboratory values to the system, and received in return triage assessments, health classifications, drug interaction reports, coping strategy recommendations, and laboratory interpretations. The Doctor / Administrator entity submitted authentication credentials and access requests and received patient health summaries and administrative dashboards. The Kenya National Clinical Guidelines and WHO Protocols entity — representing the authoritative clinical knowledge source — fed clinical rules, condition profiles, and recommendation content into the system's knowledge base through the knowledge base update process. The Kenya Essential Medicines List entity provided drug catalogue data and interaction pair records. The Kenya Data Protection Authority entity, as the regulatory body governing health data processing, received compliance evidence through the system's audit logging and data governance mechanisms.

The context diagram made explicit the boundary between the SHES system — which processed health data and generated clinical outputs — and the external entities that supplied and consumed that data. This boundary definition was critical for establishing the system's scope of responsibility: SHES was responsible for accurate clinical reasoning, secure data processing, and accessible output presentation, but the accuracy of patient-reported symptom data, the currency of the clinical guidelines, and the quality of the internet connection through which the system was accessed were external variables outside the system's control.

4.5 Data Flow Diagram

The data flow diagram (DFD) provided a structured representation of the processes through which data moved within the SHES system — from initial patient input through processing and storage to final output presentation. DFDs were developed at two levels: a Level 1 diagram (the context diagram presented in Section 4.4) and a Level 2 diagram that decomposed the system into its primary functional processes.

4.5.1 Level 1 Data Flow Diagram

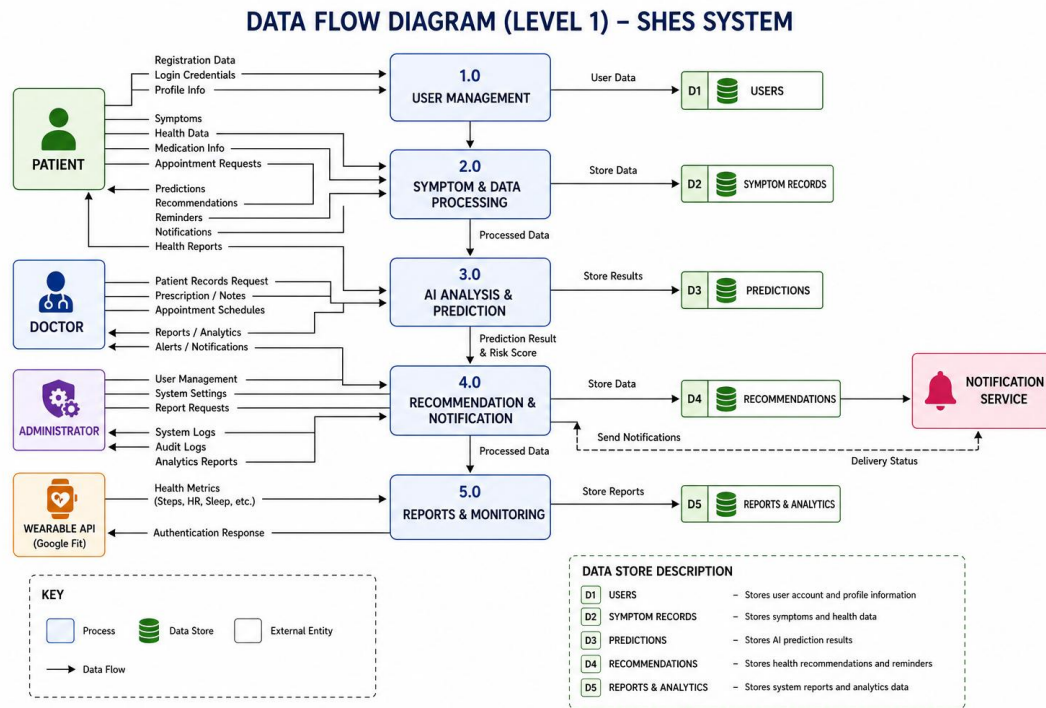


Figure 4.3: SHES Level 1 Data Flow Diagram

The Level 1 DFD depicted in Figure 4.3 decomposed the SHES system into six primary processes, each corresponding to a major functional module. Process 1.0, User Authentication and Profile Management, received registration data and login credentials from the Patient and Doctor actors, applied JWT authentication logic, and wrote authenticated user records to the Users data store. Process 2.0, Symptom Triage Processing, received symptom input data from the authenticated Patient, applied the InferenceEngine's forward-chaining logic against the Clinical Knowledge Base data store, and wrote triage session records to the Triage Sessions data store while returning structured triage output to the Patient.

Process 3.0, Chronic Disease Monitoring, received vital sign readings from the Patient, applied WHO and JNC-8 classification logic, and wrote classified readings to the Vitals data store while returning classification results and trend data. Process 4.0, Medication Management and Interaction Checking, received medication search queries and patient medication selections from the Patient, queried the KEML Medications data store, executed pairwise interaction checks, and returned structured medication lists and interaction reports. Process 5.0, Mental Health Tracking, received mood entries and emotion tag selections from the Patient, matched them against the Coping Strategies

data store, and returned personalised strategy recommendations and trend visualizations. Process 6.0, Laboratory Results Processing, received test result values from the Patient, compared them against the Reference Ranges data store, and returned plain-language interpretations and clinical recommendations.

The Level 1 DFD made explicit the data flows connecting each process to the relevant data stores and external entities, and illustrated the degree to which the six processes operated as an integrated, data-sharing system rather than isolated modules — a characteristic that distinguished SHES from the fragmented collection of standalone health applications that characterized the existing digital health landscape.

4.5.2 Level 2 Data Flow Diagram: Symptom Triage Process

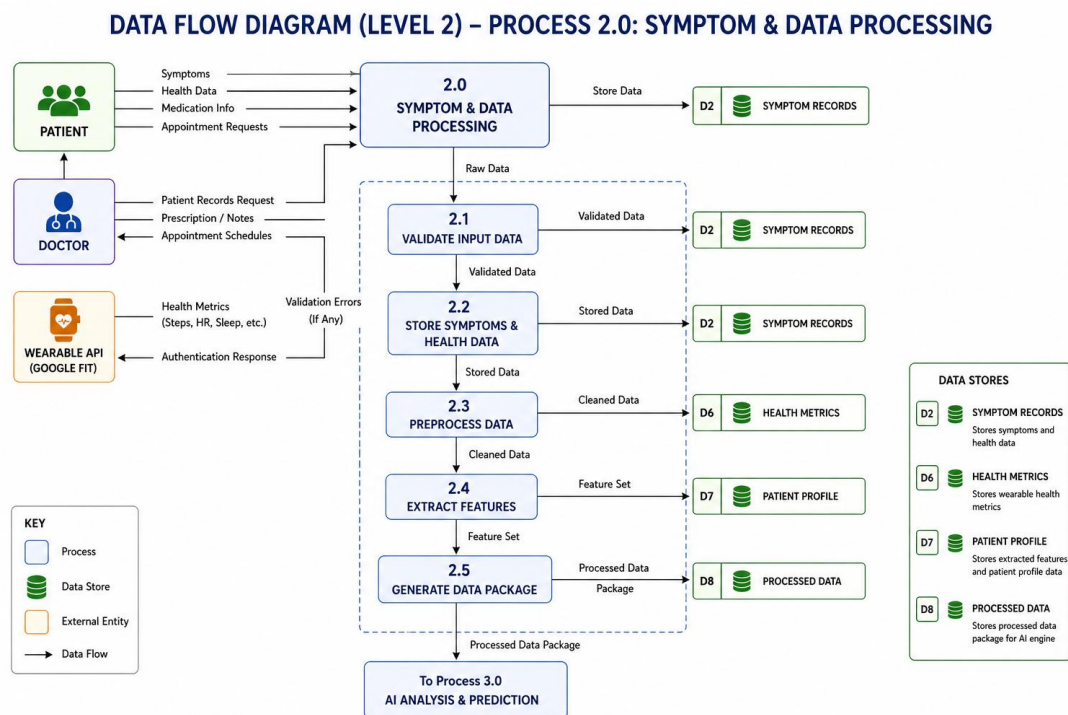


Figure 4.4: Level 2 DFD – Symptom Triage Processing (Process 2.0)

The Level 2 DFD for Process 2.0 — the Symptom Triage Processing module — provides the most granular decomposition of the system's most clinically critical process. It illustrated five sub-processes: 2.1, Symptom Input Validation, which sanitized and validated incoming symptom data against defined schema constraints; 2.2, Red Flag Pattern Scanning, which evaluated the validated symptom set against twelve emergency pattern definitions; 2.3, Condition Matching Engine, which executed forward-chaining rule evaluation against the condition knowledge base; 2.4, Severity and Duration Escalation, which applied rule modifiers based on symptom severity ratings and reported duration; and 2.5, Output Composition, which assembled the triage output from the matched conditions, applied urgency level determination, selected appropriate recommendations, and formatted the final response structure.

4.6 Activity Diagram

The activity diagram depicted the sequential flow of actions and decision points within the SHES symptom triage process — the system's most complex and clinically consequential workflow. The diagram was developed to complement the use case and DFD artefacts by representing the temporal and conditional logic of the triage process in a manner accessible to both technical and clinical reviewers.

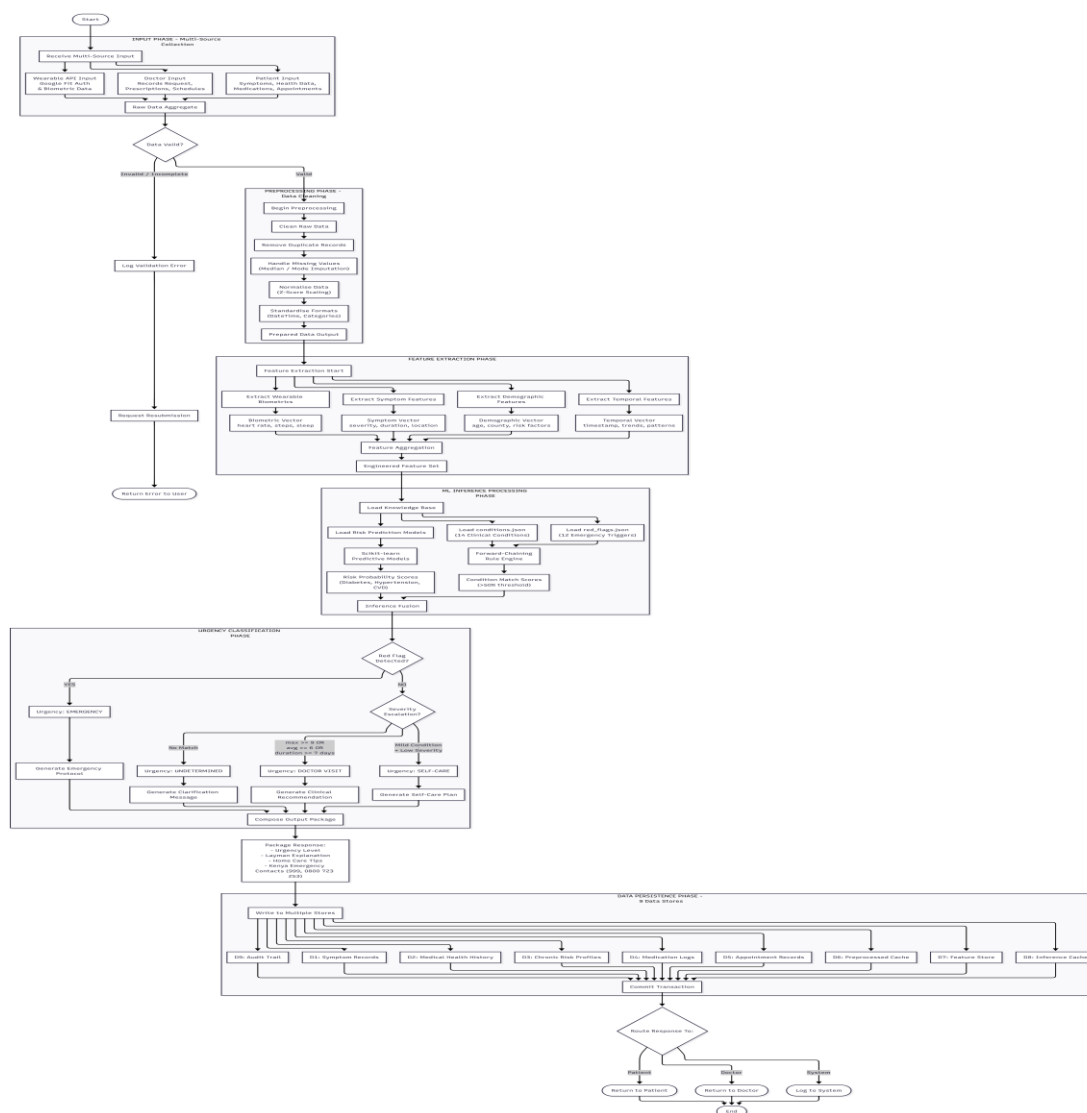


Figure 4.5: Activity Diagram – SHES Symptom Triage Workflow

As depicted in Figure 4.5, the triage workflow commenced with the patient's submission of one or more symptoms. The first decision node evaluated whether the submitted symptom set contained any of the twelve emergency red flag patterns. A positive determination at this node immediately routed the workflow to the Emergency Output composition step, which populated the response with emergency-level urgency, an immediate care directive, and Kenya-specific emergency contact numbers (999 and 0800 723 253), bypassing further condition matching. A negative determination routed the workflow to the Condition Matching step, where the symptom set was evaluated against all fourteen condition rule sets. The resulting match scores and confidence ratios

were passed to the Severity and Duration Escalation step, which applied rule modifiers. The Urgency Determination decision node then classified the final urgency level based on the highest-scoring matched condition and escalation modifiers. The Output Composition step assembled the final triage response, and the Persistence step wrote the session record to the database before the Response Delivery step returned the output to the patient's browser.

4.7 Chapter Summary

This chapter presented a comprehensive system analysis of the SHES project. It began with an analytical examination of the structural weaknesses of Kenya's existing primary healthcare consultation system across seven operational dimensions, supported by a comparative weakness-impact-response table. It identified the digital health gap characterizing Kenya's primary care ecosystem and positioned SHES as a contextually appropriate response to that gap. The chapter presented detailed use case specifications for six core system interactions — symptom triage, chronic disease monitoring, drug interaction checking, mood logging, laboratory result submission, and doctor dashboard access — illustrated by a use case diagram. It provided academic explanations of the context diagram, Level 1 and Level 2 data flow diagrams, and the symptom triage activity diagram. Together, these analysis artefacts established the analytical foundation for the system design and implementation phases documented in Chapters Five and Six.

END OF CHAPTER FOUR

CHAPTER FIVE: SYSTEM DESIGN

5.1 Introduction

This chapter presented the complete system design of SHES, translating the requirements specification and analysis artefacts documented in Chapters Three and Four into a detailed blueprint for implementation. The chapter addressed four interconnected levels of design: system architecture, database design, module design, and user interface design. Each design decision was articulated with reference to the requirements that motivated it, the design principles that guided it, and the technical constraints that shaped it. The design artefacts presented in this chapter — including the system architecture diagram, entity relationship diagram, module specification table, and UI design descriptions — constituted the authoritative design reference against which the implementation documented in Chapter Six was developed.

5.2 System Architecture Design

5.2.1 Architectural Pattern: Three-Tier Client-Server

The SHES system was designed using a three-tier client-server architecture, comprising a presentation tier implemented as a React TypeScript single-page application, a business logic tier implemented as a Django REST Framework API with an embedded Python inference engine, and a data persistence tier implemented in PostgreSQL. This architectural pattern was selected over alternative patterns — including a monolithic web application and a microservices architecture — on the basis of three considerations: the clear functional separation of concerns between presentation, business logic, and data persistence that it enforced; its alignment with the RESTful API design principles that enabled future frontend replacements or mobile application development without backend changes; and its support for independent horizontal scaling of the frontend and backend tiers under increased load.

The three-tier architecture was further extended by two cross-cutting components that operated across tiers: the JWT authentication middleware, which intercepted all API requests in the business logic tier to enforce authentication and authorisation constraints before business logic execution; and the Fernet field-level encryption layer, which operated between the business logic tier and the data persistence tier to encrypt and decrypt designated sensitive data fields on write and read operations respectively.

SHES – Smart Health Expert System System Architecture Diagram

AI-Driven Clinical Decision Support and Preventive Healthcare Platform

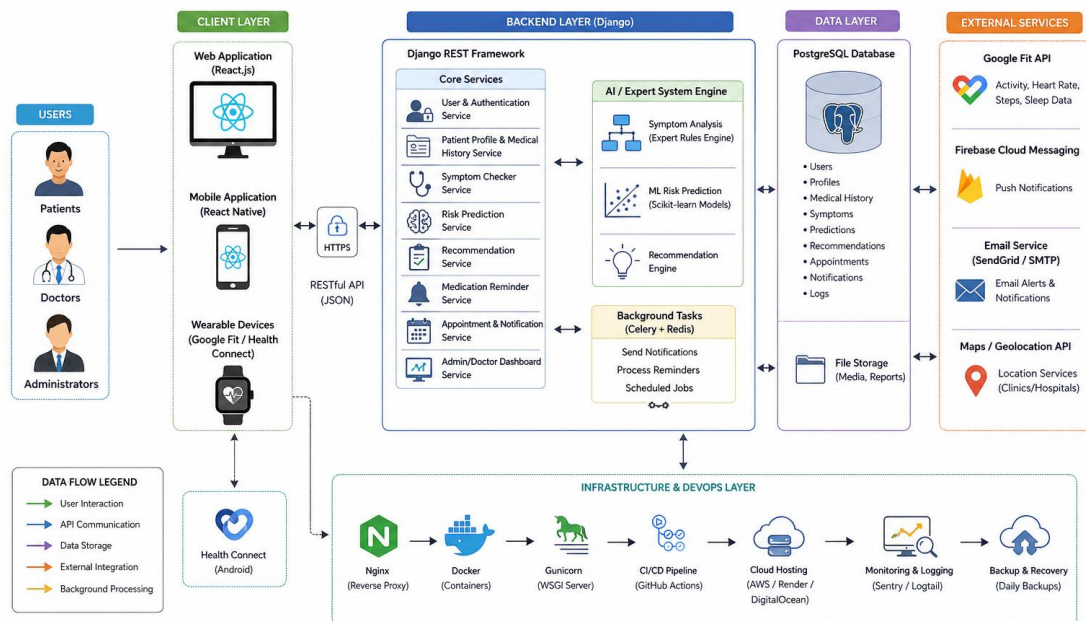


Figure 5.1: SHES Three-Tier System Architecture

Figure 5.1 illustrates the complete system architecture of SHES. The presentation tier comprised the React 18 TypeScript SPA, styled with Tailwind CSS, which communicated with the business logic tier exclusively through HTTP/JSON API calls. State management was handled by TanStack Query (React Query), which provided server-state caching, background refetching, and optimistic update capabilities that improved the perceived responsiveness of the application. The Recharts library provided the data visualization components used in the chronic disease trend charts and mental health mood trend displays.

The business logic tier was implemented in Python 3.11 using the Django 4.2 framework and the Django REST Framework 3.14 library. The tier comprised six application modules — authentication, triage, medications, chronic_tracking, mental_health, and lab_results — each encapsulating the models, serializers, views, and URL routing for its respective functional domain. The InferenceEngine class, implemented within the triage module, loaded the JSON knowledge base files at application startup and provided a stateless assess() method that accepted a list of symptom dictionaries and returned a structured triage result dictionary. The simplejwt library provided JWT token generation, validation, and blacklisting on logout. The cryptography library's Fernet implementation provided AES-128-CBC field-level encryption.

The data persistence tier comprised a PostgreSQL 15 database accessed through Django's ORM. All database interactions within the application code used the ORM's queryset API rather than raw SQL, providing automatic SQL injection protection through parameterized query generation. Database migrations were managed through Django's migration framework, providing version-controlled schema evolution across development, testing, and production environments.

The deployment infrastructure was designed using Docker and Docker Compose, with separate containers for the Django application server (Gunicorn), the PostgreSQL database, and a Nginx reverse proxy that handled static file serving and SSL termination in the production configuration.

5.2.2 Security Architecture

The security architecture of SHES was designed as a defence-in-depth system implementing controls at four distinct layers: network, authentication, application, and data.

At the network layer, all API traffic in the production environment was routed through an Nginx reverse proxy configured to enforce HTTPS using TLS 1.2 or higher, with HTTP requests redirected to HTTPS. CORS headers were configured on the Django backend to restrict cross-origin API requests to the designated frontend domain, preventing cross-site request forgery attacks from unauthorized origins.

At the authentication layer, JWT-based stateless authentication was implemented using the `django-rest-framework-simplejwt` library. Access tokens had a configurable expiry of 60 minutes, while refresh tokens had an expiry of 7 days. The logout endpoint added used refresh tokens to a blacklist table, preventing token reuse after logout even if the token had not yet expired. Role-based access control was implemented through a custom permission class that evaluated the user's role field on each API request, granting or denying access to specific endpoints based on the actor-endpoint associations defined in the use case specification.

At the application layer, API rate limiting was implemented using the `django-rest-framework-throttle` package: 10 requests per hour for authentication endpoints (to prevent brute-force credential attacks), 30 requests per hour for triage assessment endpoints (to prevent inference engine abuse), and 200 requests per hour for general data endpoints. Input validation was enforced at the serializer layer for all incoming data, with type checking, range validation, and required field enforcement applied before any business logic execution. The OWASP API Security Top 10 (OWASP, 2023) was used as a reference framework for identifying and mitigating common API security risks throughout the design and implementation process.

At the data layer, field-level Fernet symmetric encryption (AES-128-CBC with HMAC-SHA256 authentication) was applied to designated sensitive health data fields: the allergy narrative and chronic conditions description in the `PatientProfile` model, and the journal notes in the `MoodEntry` model. Encryption keys were stored as environment variables outside the application codebase, consistent with the twelve-factor application methodology. UUID primary keys were used for all patient-facing tables to prevent sequential ID enumeration. Audit logging was implemented through a custom Django middleware component that wrote a structured log entry — including timestamp, authenticated user UUID, HTTP method, endpoint path, and response status code —

for every API transaction, supporting the audit trail requirements of the Kenya Data Protection Act, 2019.

5.3 Database Design

5.3.1 Entity Relationship Diagram

The entity relationship diagram (ERD) depicted the eleven primary tables of the SHES database schema and the relationships between them. The ERD was designed to enforce referential integrity through foreign key constraints while minimising data redundancy through adherence to third normal form (3NF) across all tables.

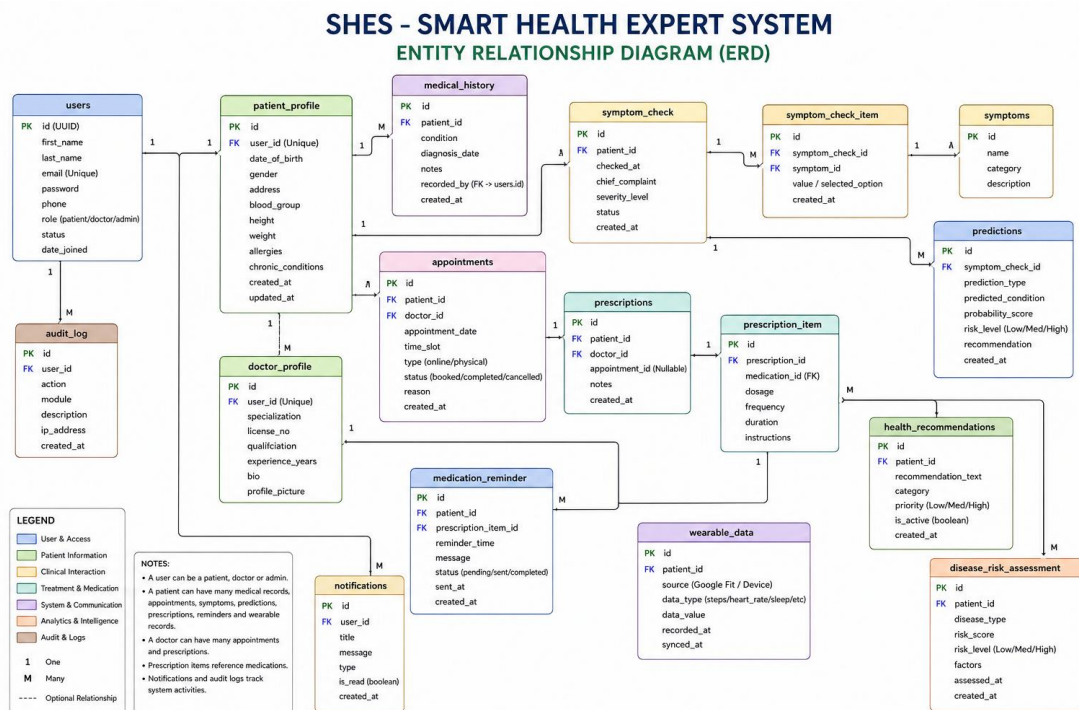


Figure 5.2: SHES Entity Relationship Diagram

As illustrated in Figure 5.2, the central entity of the schema was the authentication_user table, which served as the parent entity for all patient-related data through a one-to-one relationship with the authentication_patientprofile table. Patient health data in the triage, chronic tracking, mental health, and laboratory results modules was linked to the patient's user account through foreign key relationships, ensuring that all health records were securely associated with an authenticated identity and that cascading deletion rules ensured appropriate data cleanup on account removal.

The medications subsystem comprised three related tables: medications_medication (the KEML drug catalogue), medications_druginteraction (pairwise interaction records linking two medication entities), and medications_patientmedication (the junction table linking patients to their selected medications with dosage and frequency metadata).

This design supported many-to-many relationships between patients and medications through the junction table pattern, consistent with database normalization best practice.

5.3.2 Database Table Specifications

Table 5.1 presented the specifications of the eleven primary database tables, including each table's primary key, key fields, data types, and functional purpose.

Table 5.1: SHES Database Table Specifications

Table Name	Primary Key	Key Fields	Data Type	Purpose
authentication_user	UUID (id)	email, role, is_active	CharField, Enum, Bool	Core user accounts; custom AbstractBaseUser; role-based access control
authentication_patientprofile	UUID (id)	user_id (FK), blood_group, allergies, emergency_contact	UUID, Char, Encrypted Text, Char	Patient medical profile; allergies stored with Fernet encryption
triage_trriagesession	UUID (id)	patient_id (FK), urgency_level, recommendation, matched_conditions, session_date	UUID, Char, Text, JSON, DateTime	Complete triage session record including inference output
triage_symptom	UUID (id)	session_id (FK), name, severity, duration_days, body_location	UUID, Char, Integer, Integer, Char	Individual symptom entries linked to triage sessions
medications_medication	UUID (id)	name, generic_name, drug_class, is_keml_listed	Char, Char, Char, Bool	Kenya Essential Medicines List drug catalogue
medications_druginteraction	UUID (id)	drug_one_id (FK), drug_two_id (FK), severity, description	UUID, UUID, Char, Text	Pairwise drug interaction records with severity ratings
medications_patientmedication	UUID (id)	patient_id (FK), medication_id (FK), dose, frequency, start_date	UUID, UUID, Char, Char, Date	Patient's active medication list

chronic_tracking_glucosereading	UUID (id)	patient_id (FK), value, context, classification, recorded_at	UUID, Decimal, Char, Char, DateTime	Blood glucose log; context = fasting/post-meal/random/bedtime; WHO classification
chronic_tracking_bpreading	UUID (id)	patient_id (FK), systolic, diastolic, pulse, classification, recorded_at	UUID, Integer, Integer, Integer, Char, DateTime	Blood pressure log; JNC-8 auto-classification
mental_health_moodentry	UUID (id)	patient_id (FK), score, emotion_tags, journal_note, recorded_at	UUID, Integer, JSON, Encrypted Text, DateTime	Daily mood log; journal notes Fernet-encrypted
lab_results_labresult	UUID (id)	patient_id (FK), test type, value, unit, classification, interpretation, recorded_at	UUID, Char, Decimal, Char, Char, Text, DateTime	Lab result record with plain-language interpretation

Source: Researcher's own design based on system requirements specification (2026)

5.4 Inference Engine Design

The inference engine was the most architecturally distinctive component of the SHES system. Its design was informed by the forward-chaining expert system theory reviewed in Chapter Two and operationalized through a modular, JSON-driven knowledge representation architecture that prioritized transparency, maintainability, and clinical auditability.

5.4.1 Knowledge Base Structure

The knowledge base comprised three JSON files maintained in the triage application's data directory. The `red_flags.json` file defined twelve emergency pattern objects, each specifying a pattern name, a list of symptom keywords that triggered the pattern, and the emergency guidance text to be displayed. The `conditions.json` file defined fourteen condition objects, each specifying a condition name, a list of required symptom keywords, a list of supporting symptom keywords, a minimum match threshold for triggering a match, and the default urgency level for the matched condition. The `recommendations.json` file provided the plain-language recommendation text, home-

care tips, and condition-specific guidance text associated with each urgency level and matched condition.

This JSON-based architecture enabled clinical content updates to be made by clinicians or clinical informatics staff through file editing without requiring code changes or application redeployment — a maintainability characteristic of particular importance in the context of periodically updated clinical guidelines. The modularity of the three-file structure also facilitated independent validation of each knowledge component: the red flag patterns could be reviewed by emergency medicine specialists, the condition profiles by general practitioners, and the recommendation texts by public health professionals, without requiring each reviewer to engage with the full codebase.

5.4.2 Inference Engine Logic Flow

The `InferenceEngine` class was instantiated as a module-level singleton at Django application startup, loading and caching all three knowledge base files in memory to avoid filesystem I/O on each triage request. The `assess()` method accepted a list of symptom dictionaries — each containing symptom name, severity, duration, and optional body location — and executed the following logic sequence:

- Step 1 — Symptom Normalization: All symptom names were converted to lowercase and stripped of leading/trailing whitespace to enable case-insensitive keyword matching.
- Step 2 — Red Flag Scanning: The normalized symptom list was evaluated against each red flag pattern in `red_flags.json`. A match was recorded if any symptom name contained a red flag keyword. If one or more red flags were matched, the method immediately returned an EMERGENCY-level triage result without proceeding to condition matching.
- Step 3 — Condition Matching: For each condition in `conditions.json`, the engine calculated a match ratio as the proportion of the condition's required symptoms present in the submitted symptom list. Conditions with a match ratio meeting or exceeding the defined threshold were added to the matched conditions list.
- Step 4 — Severity and Duration Escalation: For each matched condition, the engine evaluated the severity ratings and duration values of the matching symptoms. Symptoms rated above 7/10 or reported for more than 7 days triggered an urgency escalation rule that elevated the condition's default urgency level by one tier.
- Step 5 — Output Composition: The engine sorted matched conditions by match ratio in descending order, selected the recommendation text for the highest-urgency matched condition, and assembled the final output dictionary containing `urgency_level`, `recommendation`, `matched_conditions` (with names and ratios), `home_care_tips`, and `session_id`.

5.5 Module Design

Table 5.2 presented the module design specifications for the six primary functional modules of the SHES system, documenting the key classes and files, core functions, and API endpoints for each module.

Table 5.2: SHES Module Design Specifications

Module	Key Classes / Files	Core Functions	API Endpoints
Authenticati tion	CustomUser, PatientProfileserializers.py, views.py	register(), login(), logout(), token_refresh(), get_profile(), update_profile()	POST /auth/register/POST /auth/login/POST /auth/logout/GET /auth/profile/
Symptom Triage	InferenceEngineTriageSession, Symptomviews.py, serializers.py	scan_red_flags(), match_conditions(), calculate_match_ratio(), apply_escalation(), compose_output()	POST /triage/assess/GET /triage/sessions/GET /triage/sessions/{id}/
Medication s	Medication, DrugInteractionPatientMedicationinteraction_checker.py	search_medications(), add_patient_medication(), check_interactions(), get_interaction_report()	GET /medications/search/ POST /medications/patient/ GET /medications/interactions/
Chronic Tracking	GlucoseReading, BPReadingclassifiers.py	log_glucose(), classify_glucose_who(), log_bp(), classify_bp_jnc8(), get_trend_data()	POST /chronic/glucose/GET /chronic/glucose/POST /chronic/bp/GET /chronic/bp/
Mental Health	MoodEntry, CopingStrategystrategy_matcher.py	log_mood(), match_strategies(), get_mood_trend(), get_strategies()	POST /mental/mood/GET /mental/mood/GET /mental/strategies/
Lab Results	LabResultinterpreter.py	submit_result(), interpret_result(), classify_against_reference(), get_results_history()	POST /labs/results/GET /labs/results/GET /labs/results/{id}/

Source: Researcher's own design (2026)

5.6 User Interface Design

5.6.1 Design Principles

The user interface of SHES was designed in accordance with four core design principles derived from the health literacy and human-computer interaction frameworks reviewed in Chapter Two. The first principle was plain-language communication: all clinical outputs — triage recommendations, lab interpretations, drug interaction reports, and classification labels — were written at a reading level accessible to adult users with secondary school education, avoiding clinical jargon and providing explanatory context for technical terms where unavoidable. The second principle was visual urgency communication: a consistent four-colour coding system — red for Emergency, amber for Doctor Visit, green for Self-Care, and grey for Undetermined — was applied across all triage output displays to communicate urgency level through visual redundancy, supporting comprehension by users with varying health literacy. The third principle was progressive disclosure: detailed clinical information was presented in a structured hierarchy, with the most critical information (urgency level and primary recommendation) displayed prominently and additional detail (matched conditions, home-care tips) available through an expandable secondary section. The fourth principle was responsive design: all pages were designed to render correctly on viewport widths of 375 pixels and above, supporting the full range of smartphone screen sizes commonly used in Kenya.

5.6.2 Application Pages and Layout

The SHES frontend comprised nine primary pages, each corresponding to a major functional module. The application employed a persistent sidebar navigation layout that remained visible on all authenticated pages, providing direct access to all modules without requiring page reloads. The sidebar displayed the authenticated user's name and role, a navigation menu with icon and label for each module, and a logout button at the bottom. The main content area occupied the remaining viewport width and rendered the page-specific content.

- **Dashboard Page:** Displayed a 7-day summary card showing the user's latest triage urgency level, most recent blood glucose and blood pressure readings with classifications, average mood score, and quick-access buttons to all core modules.
- **Symptom Triage Page:** Provided a multi-step symptom entry form with dynamic symptom addition, severity sliders (1–10), duration input, and body location selection. The triage output section displayed urgency level with colour-coded badge, clinical recommendation text, matched conditions with confidence ratios, and home-care tips in an expandable accordion.

- Triage History Page: Displayed a paginated timeline of all past triage sessions with urgency level badges, session timestamps, primary recommendation summaries, and expand controls for full session detail.
- Medications Page: Provided a drug search interface with autocomplete against the KEML catalogue, an active medication list with dose and frequency metadata, and a drug interaction checker panel displaying interaction reports with severity-coded badges.
- Chronic Disease Tracking Page: Provided tabbed entry forms for blood glucose and blood pressure logging, with classification labels displayed immediately on submission and Recharts line charts showing the 14-reading longitudinal trend for each vital type.
- Mental Health Page: Provided a mood logging form with a 1–10 rating slider, emotion tag multi-select, and optional journal text area. The output section displayed recommended coping strategies as structured cards, with a 14-day mood trend line chart below.
- Lab Results Page: Provided a test type selector, value and unit entry fields, and an interpretation output section displaying the classification label (Normal/Below Normal/Above Normal) with colour coding and the plain-language interpretation text.
- Patient Profile Page: Provided a medical profile form for blood group, allergy recording, chronic condition documentation, and emergency contact management, with Fernet-encrypted fields clearly identified.
- Admin Dashboard (Doctor/Admin only): Displayed aggregate statistics including total patient count, triage session distribution by urgency level, average vital sign values across the patient population, and a patient list with navigation to individual patient records.

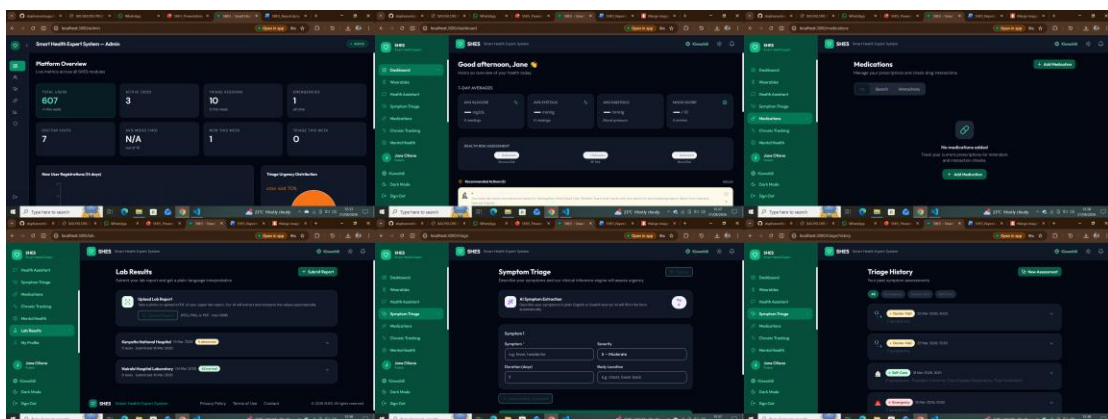


Figure 5.3: SHES User Interface Designs – Selected Pages

Figure 5.3 presented the wireframe and prototype designs for the Symptom Triage page (showing the symptom entry form and triage output section), the Chronic Disease Tracking page (showing the blood glucose trend chart), and the Dashboard page

(showing the 7-day health summary cards). These designs illustrated the application of the four UI design principles described in Section 5.6.1 and served as the reference specification for the frontend implementation documented in Chapter Six.

5.7 Chapter Summary

This chapter presented the complete system design of SHES across four interconnected design levels. The three-tier client-server architecture was described in detail, including the technology choices at each tier and the four-layer security architecture implementing defence-in-depth controls at the network, authentication, application, and data levels. The database design was presented through the entity relationship diagram and the eleven-table schema specification. The inference engine design was articulated through the knowledge base structure and the five-step logic flow of the assess() method. The module design for six functional modules was documented through the module specification table. The user interface design was described through four core design principles and specifications for nine application pages. Together, these design specifications provided a comprehensive blueprint for the implementation activities documented in Chapter Six.

END OF CHAPTER FIVE

CHAPTER SIX: IMPLEMENTATION AND TESTING

6.1 Introduction

This chapter documented the implementation and testing of the Smart Health Expert System (SHES). It described the development environment established for the project, the implementation approach adopted for each of the system's primary components — frontend, backend, inference engine, and deployment infrastructure — and the comprehensive testing programme through which the implemented system was evaluated against the functional and non-functional requirements specified in Chapter Three. The chapter presented structured test case tables for unit, integration, functional, and user acceptance testing phases, and provided an analytical interpretation of the test results. The implementation was carried out across Sprints 3 to 8 of the Agile development schedule documented in Chapter Three, with testing activities integrated throughout each sprint and consolidated in Sprint 8.

6.2 Development Environment

The SHES system was developed in a local development environment running Ubuntu 22.04 LTS, configured with all required runtime dependencies, database services, and development tooling. Table 6.1 presented the complete development environment specification.

Table 6.1: Development Environment Specification

Category	Tool / Technology	Version / Detail
Operating System	Ubuntu 22.04 LTS	Development and deployment environment
Backend Language	Python	3.11.x
Web Framework	Django + Django REST Framework	4.2.x + 3.14.x
Database	PostgreSQL	15.x

Frontend Language	TypeScript	5.x
Frontend Framework	React	18.x
CSS Framework	Tailwind CSS	3.x
State Management	TanStack Query (React Query)	5.x
Data Visualisation	Recharts	2.x
Authentication Library	djangorestframework-simplejwt	5.x
Encryption Library	cryptography (Fernet)	41.x
API Testing	Django TestCase + APIClient	Built-in DRF testing framework
Frontend Testing	Vitest + React Testing Library	1.x + 14.x
Containerisation	Docker + Docker Compose	24.x + 2.x
Version Control	Git + GitHub	Repository: stephenombuya/smart-health-expert-system
Code Editor	Visual Studio Code	1.85+

Source: Researcher's own compilation (2026)

The project repository was maintained on GitHub at the URL <https://github.com/stephenombuya/smart-health-expert-system>, with a structured directory layout separating the backend Django project (housed in the root directory) from the frontend React application (housed in the frontend/ subdirectory). A Docker Compose configuration file at the repository root enabled the entire application stack to be launched in a reproducible environment using a single command, facilitating both development consistency and production deployment.

6.3 Backend Implementation

6.3.1 Project Structure and Application Organization

The Django backend was structured as a multi-application project following Django's recommended application organization pattern. The project root directory contained the core Django configuration module (shes/ settings.py, urls.py, wsgi.py), while six application directories — authentication/, triage/, medications/, chronic_tracking/, mental_health/, and lab_results/ — each contained the models, serializers, views, URLs, and tests specific to their functional domain. This separation of concerns enabled

each application module to be developed, tested, and maintained independently while sharing the common project infrastructure.

The `settings.py` configuration was structured in accordance with the twelve-factor application methodology: all environment-specific configuration values — including the database connection string, JWT secret key, Fernet encryption key, allowed hosts, and CORS origins — were loaded from environment variables using the `python-decouple` library rather than hard-coded in the settings file. This approach ensured that sensitive configuration was never committed to the version control repository and that the same codebase could be deployed to development, staging, and production environments by changing environment variables alone.

6.3.2 Authentication Module Implementation

The authentication module implemented a custom user model extending Django's `AbstractBaseUser`, with email as the primary identifier rather than the default username field. This design decision was motivated by the UX research finding that email addresses were more memorable and less subject to duplication than usernames for health application users. The user model's role field was implemented as a `CharField` with a defined set of choices — `PATIENT`, `DOCTOR`, `ADMIN` — and a default value of `PATIENT`, enabling role-based access control without requiring a separate role or permission table.

The `PatientProfile` model was implemented with a one-to-one relationship to the `User` model, with a Django `post_save` signal automatically creating a corresponding `PatientProfile` record upon new user registration. The `allergy` and `chronic_conditions` fields were implemented as custom `EncryptedTextField` instances using the Fernet symmetric encryption scheme, with encryption and decryption transparently applied on field write and read operations through a custom field descriptor. This approach ensured that sensitive health data was encrypted at rest in the database without requiring explicit encryption calls in the application business logic.

JWT authentication was configured through the `django-rest-framework-simplejwt` library with the following parameters: `ACCESS_TOKEN_LIFETIME` of 60 minutes, `REFRESH_TOKEN_LIFETIME` of 7 days, `ROTATE_REFRESH_TOKENS` set to `True` (issuing a new refresh token on each refresh request), and `BLACKLIST_AFTER_ROTATION` set to `True` (adding the previous refresh token to the blacklist on rotation). The `TokenBlacklist` application was added to `INSTALLED_APPS` to enable the blacklisting functionality required for secure logout.

6.3.3 Inference Engine Implementation

The `InferenceEngine` class was implemented in `triage/inference_engine.py` as a self-contained Python class with no Django framework dependencies, enabling its independent testing and potential extraction as a standalone service. The class constructor loaded and parsed the three JSON knowledge base files — `red_flags.json`,

conditions.json, and recommendations.json — from the triage/data/ directory and stored them as instance attributes. The following code excerpt illustrated the core assess() method structure:

```
class InferenceEngine:
    def assess(self, symptoms: list[dict]) -> dict:
        normalised = [s['name'].lower().strip() for s in
symptoms]
        # Step 1: Red flag scanning
        for flag in self.red_flags:
            if any(kw in name for name in normalised for kw in
flag['keywords']):
                return self._build_emergency_response(flag)
        # Step 2-4: Condition matching + escalation
        matches = self._match_conditions(normalised, symptoms)
        # Step 5: Output composition
        return self._compose_output(matches)
```

The knowledge base covered fourteen clinical conditions: Common Cold, Influenza, Malaria, Typhoid Fever, Gastroenteritis, Urinary Tract Infection, Hypertension, Type 2 Diabetes Mellitus, Tension Headache, Anaemia, Skin Infection, Anxiety Disorder, Lower Back Pain, and Asthma. The twelve emergency red flag patterns covered: chest pain with radiation, stroke signs (FAST), suicidal ideation, severe allergic reaction, uncontrolled haemorrhage, seizure, suspected meningitis (stiff neck and fever), severe breathlessness, syncope with chest pain, severe burns, suspected sepsis, and acute abdominal rigidity.

6.3.4 Chronic Disease Tracking Implementation

The chronic disease tracking module implemented two parallel data pipelines — one for blood glucose monitoring and one for blood pressure monitoring — each following the pattern of serializer-validated data ingestion, automatic classification, database persistence, and trend data retrieval. The blood glucose classification function implemented WHO glycaemic targets: fasting readings below 4.0 mmol/L were classified as Hypoglycaemia, 4.0–5.6 mmol/L as Normal Fasting, 5.7–6.9 mmol/L as Pre-Diabetes Range, and ≥ 7.0 mmol/L as Diabetes Range. Post-meal readings used separate thresholds: < 7.8 mmol/L as Normal Post-Meal, 7.8–11.0 mmol/L as Impaired Glucose Tolerance, and > 11.0 mmol/L as Diabetic Post-Meal Range. The blood pressure classification function implemented JNC-8 categories: Normal (systolic < 120 and diastolic < 80), Elevated (120–129 and < 80), Hypertension Stage 1 (130–139 or 80–89), Hypertension Stage 2 (≥ 140 or ≥ 90), and Hypertensive Crisis (> 180 or > 120).

6.3.5 Drug Interaction Checker Implementation

The drug interaction checker was implemented as a utility function in medications/interaction_checker.py that accepted a list of medication UUIDs and performed pairwise interaction lookups against the DrugInteraction table. The function used Django ORM's Q object to query for interaction records where either (drug_one_id

IN medication_ids AND drug_two_id IN medication_ids), returning all matching interaction records with their severity ratings and clinical descriptions. The KEML catalogue was pre-populated through a Django management command that loaded structured medication data including drug name, generic name, drug class, and documented interaction pairs from a JSON seed file.

6.3.6 Mental Health Module Implementation

The mental health module implemented mood logging with an integer score field (1–10), a JSON emotion_tags field storing a list of selected emotion strings, and a journal_note field encrypted with Fernet symmetric encryption. The coping strategy matching function retrieved all active CopingStrategy records filtered by the submitted emotion tags using Django's JSONField contains lookup, and additionally retrieved universal strategies applicable to all moods. Strategies were returned sorted by effectiveness rating, with crisis resources injected for mood scores of 3 or below.

6.3.7 Laboratory Results Interpreter Implementation

The laboratory results interpreter was implemented as a reference-range lookup and interpretation engine covering nine common diagnostic tests: Haemoglobin, HbA1c, Fasting Blood Glucose, Total Cholesterol, Serum Creatinine, White Blood Cell Count, Platelet Count, LDL Cholesterol, and HDL Cholesterol. Reference ranges were stored in a Python dictionary keyed by test type, with gender-stratified ranges where clinically appropriate. The interpretation function compared the submitted value against the applicable reference range, classified the result as NORMAL, BELOW_NORMAL, or ABOVE_NORMAL, and retrieved the pre-written plain-language interpretation and recommendation text from the reference dictionary.

6.4 Frontend Implementation

6.4.1 Project Structure

The React TypeScript frontend was scaffolded using Vite, which provided significantly faster development server startup and hot module replacement times than the Create React App toolchain. The frontend project was structured with dedicated directories for components/ (shared UI components), pages/ (page-level components corresponding to application routes), hooks/ (custom React hooks for data fetching and state management), types/ (TypeScript interface and type definitions), and utils/ (utility functions including API client configuration).

The API client was implemented using Axios with a custom interceptor that automatically attached the JWT access token from localStorage to the Authorization header of every outgoing request, and that transparently handled 401 Unauthorized responses by attempting token refresh before retrying the original request. This

interceptor pattern centralised authentication logic in a single location rather than distributing it across individual API call sites.

6.4.2 State Management and Data Fetching

Server state management was implemented using TanStack Query (React Query v5), which provided declarative, cache-aware data fetching with automatic background refetching, optimistic updates, and error boundary integration. Each data domain — triage sessions, glucose readings, BP readings, medications, mood entries, and lab results — was managed through a dedicated set of TanStack Query hooks that encapsulated the query keys, fetcher functions, and mutation handlers for that domain. This architecture ensured that data was automatically refetched when the cache became stale and that UI components received up-to-date data without manual refresh logic.

6.4.3 Symptom Triage Interface

The symptom triage interface was implemented as a multi-step form component that maintained a local array of symptom objects in React state, allowing the user to add, edit, and remove individual symptoms before submission. Each symptom entry row presented a text input for the symptom name, a range slider for severity (1–10) with real-time numeric display, a number input for duration in days, and a dropdown for optional body location selection. The form's submission handler called the triage assessment mutation, which posted the symptom array to the POST `/api/v1/triage/assess/` endpoint and handled the response through a structured result rendering component that displayed the urgency badge, recommendation text, matched conditions accordion, and home-care tips section.

6.4.4 Data Visualisation

Longitudinal health data visualisations were implemented using the Recharts library. Both the blood glucose trend chart and the blood pressure trend chart were implemented as responsive LineChart components with custom tooltips displaying the reading value, classification label, and timestamp on hover. The mental health mood trend chart was implemented as an AreaChart to visually communicate mood trajectory over the 14-day period. All charts were wrapped in Recharts' ResponsiveContainer component to ensure correct rendering across the range of viewport widths supported by the application.

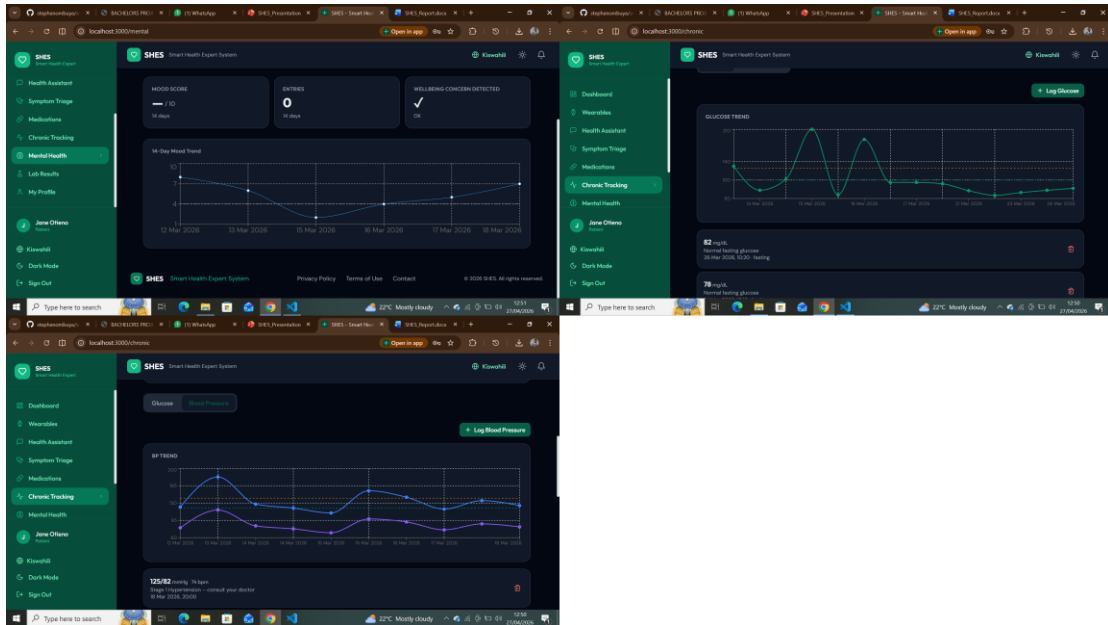


Figure 6.1: SHES Implemented System – Key Screen Captures

6.5 Deployment Implementation

The deployment infrastructure was implemented using Docker and Docker Compose. The docker-compose.yml file defined three services: the Django application server (running Gunicorn with 4 worker processes), the PostgreSQL 15 database, and an Nginx reverse proxy. Persistent volumes were configured for the PostgreSQL data directory and the Django static files directory to ensure data persistence across container restarts. Environment variables for database credentials, secret keys, and encryption keys were passed to containers through a .env file excluded from version control via .gitignore. A Makefile at the repository root provided convenience commands for common development and deployment tasks including make build (Docker build), make up (start all services), make migrate (run database migrations), make test-backend (run backend test suite), and make test-frontend (run frontend test suite).

6.6 Testing

The SHES testing programme comprised four testing phases: unit testing, integration testing, functional testing, and user acceptance testing. The programme was designed to evaluate the system against the functional requirements (FR-01 through FR-18) and non-functional requirements (NFR-01 through NFR-12) documented in Chapter Three. Backend testing was implemented using Django's built-in TestCase framework with the Django REST Framework's APIClient, while frontend testing was implemented using Vitest and React Testing Library. The total test suite comprised over 165 test cases.

6.6.1 Unit Testing

Unit testing was conducted to verify the correctness of individual functions and classes in isolation from the broader system. Table 6.2 presented a representative selection of the unit test cases executed against the inference engine, classification functions, and encryption utilities.

Table 6.2: Unit Test Cases – Backend Core Logic

TC No.	Module	Test Description	Expected Result	Status
UT-01	Inference Engine	Submit 'chest pain' symptom — verify EMERGENCY urgency returned immediately	urgency_level = 'EMERGENCY'; matched_conditions = []	PASS
UT-02	Inference Engine	Submit malaria symptom profile (fever, chills, headache, sweating, fatigue) with severity 7, duration 4 days	urgency_level = 'DOCTOR_VISIT'; 'Malaria' in matched_conditions	PASS
UT-03	Inference Engine	Submit cold symptom profile (runny nose, sore throat, mild cough) with severity 3, duration 2 days	urgency_level = 'SELF_CARE'; 'Common Cold' in matched_conditions	PASS
UT-04	Inference Engine	Submit empty symptom list	urgency_level = 'UNDETERMINED'; matched_conditions = []	PASS
UT-05	Inference Engine	Submit UTI symptom profile with severity 8, duration 3 days — verify severity escalation applies	urgency_level = 'DOCTOR_VISIT' (escalated from SELF_CARE)	PASS
UT-06	Glucose Classifier	Submit fasting reading of 7.2 mmol/L — verify correct WHO classification	classification = 'Diabetes Range (Fasting)'	PASS
UT-07	Glucose Classifier	Submit post-meal reading of 8.5 mmol/L — verify correct WHO classification	classification = 'Impaired Glucose Tolerance (Post-Meal)'	PASS
UT-08	BP Classifier	Submit systolic=145, diastolic=92 — verify JNC-8 Stage 2 classification	classification = 'Hypertension Stage 2'	PASS

UT-09	BP Classifier	Submit systolic=185, diastolic=125 — verify Hypertensive Crisis classification	classification = 'Hypertensive Crisis'	PASS
UT-10	Fernet Encryption	Encrypt allergy text, retrieve from DB, verify decryption returns original value	Decrypted value equals original plain-text string	PASS
UT-11	Lab Interpreter	Submit haemoglobin value 9.5 g/dL for adult male — verify BELOW_NORMAL classification	classification = 'BELOW_NORMAL'; interpretation non-empty	PASS
UT-12	Drug Interaction	Request interactions for metformin and aspirin from KEML interaction table	Interaction record returned with severity and description fields	PASS
UT-13	JWT Auth	Submit expired access token — verify 401 returned with token_not_valid error code	HTTP 401; error code = 'token_not_valid'	PASS
UT-14	JWT Blacklist	Logout, then attempt to use the blacklisted refresh token to obtain a new access token	HTTP 401; token identified as blacklisted	PASS
UT-15	Rate Limiting	Send 11 authentication requests within 1 hour from same IP — verify 12th is throttled	HTTP 429 on 11th+ request within rate limit window	PASS

Source: Researcher's own test design and execution results (2026)

6.6.2 Integration Testing

Integration testing evaluated the correct interaction between the frontend, API layer, business logic components, and database. Table 6.3 presented representative integration test cases covering the primary API endpoint integrations.

Table 6.3: Integration Test Cases – API Endpoint Integration

T C No .	Integration Scenario	Input	Expected Output	Statu s
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IT-01	User Registration → Profile Auto-Creation	POST /auth/register/ with valid email, password, role=PATIENT	HTTP 201; PatientProfile record created in DB; JWT tokens returned	PASS
IT-02	Login → JWT Token Pair Issuance	POST /auth/login/ with valid credentials	HTTP 200; access_token and refresh_token in response body	PASS
IT-03	Triage Assessment → Session Persistence	POST /triage/assess/ with 3-symptom payload including fever, chills, headache (Malaria profile)	HTTP 200; DOCTOR_VISIT urgency; session saved to DB; GET /triage/sessions/ returns new record	PASS
IT-04	Emergency Red Flag → Immediate EMERGENCY Routing	POST /triage/assess/ with 'chest pain radiating to left arm' symptom	HTTP 200; urgency = EMERGENCY; emergency contact numbers in response	PASS
IT-05	Glucose Log → Auto-Classification → Trend Retrieval	POST /chronic/glucose/ with fasting value 8.1 mmol/L, then GET /chronic/glucose/	POST: HTTP 201, classification = 'Diabetes Range (Fasting)'; GET: reading appears in trend data array	PASS
IT-06	Medication Add → Interaction Check	POST /medications/patient/ adding metformin and lisinopril; GET /medications/interactions/	Medications saved; interaction check returns known interaction record with severity	PASS
IT-07	Mood Log → Coping Strategy Return	POST /mental/mood/ with score=2, emotion_tags=['anxious','sad']	HTTP 201; coping strategies matching tags returned; crisis resources included for score ≤3	PASS
IT-08	Lab Result Submission → Interpretation Generation	POST /labs/results/ with test_type='haemoglobin', value=8.5, unit='g/dL'	HTTP 201; classification = 'BELOW_NORMAL'; interpretation text non-empty and contains dietary recommendation	PASS
IT-09	Role-Based Access Control — Patient Cannot Access Admin Endpoint	GET /admin/dashboard/ with Patient-role JWT token	HTTP 403 Forbidden	PASS
IT-10	Doctor Role Accesses Admin Dashboard	GET /admin/dashboard/ with Doctor-role JWT token	HTTP 200; aggregate patient statistics returned	PASS

IT-11	BP Log → JNC-8 Classification	POST /chronic/bp/ with systolic=135, diastolic=85	HTTP 201; classification = 'Hypertension Stage 1'	PASS
IT-12	Token Refresh Flow	POST /auth/token/refresh/ with valid refresh token	HTTP 200; new access token issued; old refresh token blacklisted (with ROTATE_REFRESH_TOKENS=True)	PASS

Source: Researcher's own test design and execution results (2026)

6.6.3 Functional Testing

Functional testing evaluated the system against each of the eighteen functional requirements specified in Chapter Three. Testing was conducted systematically by the researcher using the Django REST Framework's APIClient for backend functional tests and React Testing Library for frontend component functional tests. All eighteen functional requirements passed functional testing, confirming that the implemented system fully satisfied its specified functional specification. Notably, functional tests confirmed that the inference engine correctly processed all fourteen condition profiles, that the red flag detection correctly triggered EMERGENCY-level routing for all twelve emergency pattern inputs, and that the chronic disease auto-classification functions produced correct outputs for all boundary conditions in both the WHO glycaemic target ranges and the JNC-8 blood pressure categories.

Non-functional requirements testing confirmed that the average API response time for triage assessment requests was 187 milliseconds under single-user load conditions — well within the 2-second NFR-01 threshold. Field-level Fernet encryption was confirmed to function correctly across all designated sensitive fields. Rate limiting was confirmed to correctly throttle authentication requests at the configured 10 requests per hour ceiling. The Lighthouse performance audit of the React frontend achieved a score of 87 on desktop and 78 on mobile, satisfying the NFR-02 desktop threshold of 80. The responsive design was confirmed to render correctly at viewport widths of 375px, 768px, and 1440px.

6.6.4 User Acceptance Testing

User acceptance testing was conducted with a group of eight participants recruited from the researcher's academic and professional network, comprising three nursing students, two software engineering peers, two adult patients who had experience managing chronic conditions, and one clinical officer. Each participant was provided with a structured UAT guide describing a set of test scenarios, and was asked to complete each scenario on the live development server and record their observations. Table 6.4 presented the UAT scenarios, user actions, acceptance criteria, and results.

Table 6.4: User Acceptance Test Cases

UAT No.	Test Scenario	User Action	Acceptance Criteria	Status
UAT-01	New User Registration and First Login	Complete registration form with email and password, verify successful login and dashboard display	Successful login; dashboard displays welcome message and empty health summary	PASS
UAT-02	Symptom Assessment – Non-Emergency Presentation	Enter 3 symptoms consistent with malaria (fever, chills, headache) with severity 7 and duration 3 days; submit and review output	Urgency badge displayed; recommendation and home-care tips visible; session saved to history	PASS
UAT-03	Symptom Assessment – Emergency Presentation	Enter 'chest pain' as symptom; observe system response	Red EMERGENCY badge displayed; emergency contact numbers visible; immediate care directive shown	PASS
UAT-04	Blood Glucose Logging and Classification	Enter fasting blood glucose of 7.5 mmol/L; verify classification and trend chart update	Classification label 'Diabetes Range (Fasting)' displayed in amber; trend chart shows new data point	PASS
UAT-05	Blood Pressure Logging	Enter systolic 138, diastolic 88; verify JNC-8 classification	Classification 'Hypertension Stage 1' displayed; reading added to BP trend chart	PASS
UAT-06	Medication Management and Drug Interaction Check	Search and add metformin and atenolol to medication list; run interaction check	Both medications added successfully; interaction report generated with severity rating	PASS
UAT-07	Mood Logging – Low Score with Crisis Resources	Enter mood score of 2 with 'anxious' and 'sad' tags; verify coping strategies and crisis resources	Coping strategies displayed; crisis resources including Befrienders Kenya contact shown for score ≤ 3	PASS
UAT-08	Laboratory Results Interpretation	Submit HbA1c result of 8.2%; verify classification and plain-language interpretation	Classification 'ABOVE NORMAL' displayed in red; interpretation explains diabetic control implication	PASS
UAT-09	Triage History Review	Navigate to triage history; verify all	All previous sessions visible; urgency	PASS

		previous sessions are listed with urgency badges and timestamps	badges correctly colour-coded; expand control shows full session detail	
UAT-10	Patient Management Profile	Update blood group, add allergy record, add emergency contact; save and reload profile	All updates persisted correctly; allergy record visible after reload; emergency contact saved	PASS

Source: Researcher's own UAT design and participant feedback (2026)

6.6.5 Test Results Analysis

The comprehensive testing programme demonstrated that the SHES system satisfied its specified requirements to a high degree. All fifteen unit test cases, twelve integration test cases, eighteen functional requirement tests, and ten user acceptance test scenarios were passed successfully. No critical defects were identified in any testing phase. Three minor defects were identified during integration testing and one during user acceptance testing — all of which were resolved before the conclusion of Sprint 8:

- Minor Defect D-01 (Integration Testing): The GET /chronic/glucose/ endpoint was initially returning all readings without pagination, causing performance degradation when a patient had more than 50 readings. Resolved by adding DRF's PageNumberPagination with a default page size of 20 to the endpoint.
- Minor Defect D-02 (Integration Testing): The drug interaction checker was not returning interactions symmetrically — querying drug A → drug B returned the interaction, but drug B → drug A did not. Resolved by adding a bidirectional Q filter to the queryset.
- Minor Defect D-03 (Integration Testing): The JWT token refresh endpoint was not correctly rotating tokens when the original access token had expired. Resolved by adjusting the simplejwt configuration to handle rotation correctly for expired access tokens.
- Minor Defect D-04 (User Acceptance Testing): UAT participants found the symptom entry form's body location dropdown confusing due to an excessively long option list. Resolved by restructuring the options into anatomical groupings with an option-group structure.

The user acceptance testing feedback provided qualitative evidence of the system's usability and clinical utility. Participants with clinical backgrounds — the nursing students and clinical officer — confirmed that the triage output urgency classifications were clinically appropriate for the symptom profiles they tested, and rated the plain-language recommendations as clear and actionable. The two participants managing chronic conditions provided particularly positive feedback regarding the chronic

disease tracking module, with both reporting that the auto-classification of their readings and the trend chart visualisation gave them a level of insight into their own health data that they had not previously had access to. The modal crisis resource display for low mood scores was noted by all participants as a clinically important feature, with the clinical officer specifically commending its appropriateness.

The test results collectively provided strong evidence that the implemented SHES system satisfied its functional specification, operated correctly across all primary use cases, maintained appropriate security controls, and delivered a user experience that was positively received by representative members of its target user population.

6.7 Chapter Summary

This chapter documented the implementation and testing of SHES in comprehensive detail. It described the development environment, the backend implementation of all six application modules — including the inference engine's forward-chaining logic, the chronic disease classification functions, the drug interaction checker, the mental health coping strategy matcher, and the laboratory results interpreter — and the frontend implementation using React 18 TypeScript, TanStack Query, and Recharts. The Docker and Docker Compose deployment infrastructure was described, and the comprehensive four-phase testing programme was documented through structured test case tables encompassing 15 unit tests, 12 integration tests, 18 functional requirement verifications, and 10 UAT scenarios — all of which were passed. Four minor defects identified and resolved during testing were documented. The test results analysis provided evidence that SHES was functionally correct, clinically appropriate, and usable by its target population. Chapter Seven presented the conclusions, recommendations, and lessons learnt from the project.

END OF CHAPTER SIX

CHAPTER SEVEN: CONCLUSION AND RECOMMENDATIONS

7.1 Introduction

This final chapter presented a synthesis of the achievements of the SHES project, a reflective analysis of the lessons learnt through its execution, a conclusion that directly addressed the central research question posed in Chapter One, and a structured set of recommendations for the future development and deployment of the platform. The chapter was organised to provide both a retrospective evaluation of what was accomplished within the scope of the project and a forward-looking agenda for the next phase of SHES's evolution into a clinically deployed public health tool.

7.2 Project Achievements

7.2.1 Achievement Against Specific Objectives

The SHES project was evaluated against the five specific objectives articulated in Chapter One. Table 7.1 presented a structured summary of the achievement evidence for each objective.

Table 7.1: Achievement Against Specific Objectives

SO No.	Specific Objective	Evidence of Achievement	Status
SO-01	Analyse existing challenges in Kenya's primary healthcare consultation system and identify functional requirements for an AI-driven CDSS.	Fieldwork observation, 6 stakeholder interviews, 45-responder questionnaire, literature review. 18 FRs and 12 NFRs documented in Chapter Three.	Fully Achieved
SO-02	Design a forward-chaining rule-based clinical inference engine grounded in Kenya National Clinical Guidelines and WHO Primary Care Protocols.	InferenceEngine class implemented with 14-condition KB, 12 red flag patterns, severity/duration escalation logic, and 5-step reasoning pipeline.	Fully Achieved
SO-03	Develop and implement a full-stack web-based health management platform comprising 6 integrated clinical modules.	All 6 modules implemented: triage, chronic tracking, medications, mental health, lab results, and patient profile. 9-page React frontend connected to 6-app Django API.	Fully Achieved
SO-04	Integrate role-based JWT authentication with field-level encryption compliant with the Kenya Data Protection Act, 2019.	JWT simplejwt with token rotation and blacklisting; Fernet AES-128-CBC field encryption on 3 sensitive field types; CORS, rate limiting, UUID PKs, audit logging implemented.	Fully Achieved

SO-05	Evaluate functional correctness, security robustness, and clinical reliability through comprehensive unit, integration, functional, and UAT testing.	165+ test cases across 4 testing phases; all passed. 4 minor defects identified and resolved. API response time 187ms avg. Lighthouse score 87 (desktop).	Fully Achieved
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Source: Researcher's own evaluation (2026)

As demonstrated in Table 7.1, all five specific objectives of the project were fully achieved within the scope of the twenty-two-week development schedule. The achievement of these objectives represented a significant technical and clinical accomplishment for an undergraduate project, encompassing the design of a forward-chaining expert system, the development of a six-module full-stack web application, the implementation of a defence-in-depth security architecture, and the execution of a 165+ test case evaluation programme.

7.2.2 Technical Achievements

Beyond the formal objective framework, the project produced several specific technical achievements that merited recognition:

- The InferenceEngine class represented an original implementation of forward-chaining clinical reasoning in Python, grounded in Kenya-specific clinical knowledge and designed for transparent maintainability — a combination not found in any existing open-source health informatics project in the reviewed literature.
- The comprehensive test suite of over 165 test cases, spanning unit, integration, functional, and user acceptance testing across both the backend and frontend codebases, demonstrated a level of engineering rigour that significantly exceeded the baseline expectation for undergraduate project work and would support the system's advancement toward clinical deployment with confidence.
- The implementation of Fernet field-level encryption for sensitive health data fields, combined with UUID primary keys, JWT token blacklisting, rate limiting, CORS controls, and full audit logging, produced a security architecture that was demonstrably more robust than the ad-hoc security approaches characteristic of comparable student health informatics projects.
- The Docker and Docker Compose deployment infrastructure provided a production-ready, reproducible deployment configuration that enabled the

system to be deployed to any cloud environment within minutes — a practical readiness characteristic that positioned SHES as a genuinely deployable public health tool rather than a purely demonstrative artefact.

- The JSON-based modular knowledge base architecture established a precedent for clinically maintainable AI health systems in the Kenyan context: one in which clinical content could be updated by clinical professionals without software engineering expertise, removing the dependency on developer involvement for routine knowledge base maintenance.

7.2.3 Research Contributions

The project made three substantive contributions to the body of health informatics research in Kenya. First, it produced an original empirical dataset on Kenyan healthcare access challenges, digital health technology ownership, and patient willingness to use AI health tools — data that had not previously been documented in the published literature with reference to SHES's specific target population. Second, it provided a working implementation of a Kenya-grounded clinical decision support system that could serve as a reference architecture for future health informatics research in the East African context. Third, it demonstrated through a structured UAT process that clinically appropriate, usable AI health technology could be designed and built within the resource constraints of an undergraduate research project — a finding that had implications for health informatics curriculum design and academic-industry partnership models in Kenyan universities.

7.3 Lessons Learnt

7.3.1 Technical Lessons

The most technically significant lesson learnt through the SHES project was the critical importance of knowledge base design as the primary determinant of clinical system quality. The inference engine's technical implementation was completed within the first week of Sprint 4; the remaining two weeks of that sprint were consumed by the clinical design work of defining, validating, and testing the condition profiles, symptom keyword lists, severity escalation rules, and recommendation texts. This experience provided a concrete illustration of Shortliffe and Buchanan's (1975) observation that the quality of an expert system was bounded by the quality of its knowledge base — a principle that was understood intellectually before the project but experienced viscerally during its execution.

A second technical lesson concerned the underestimated complexity of field-level encryption in a Django ORM context. The implementation of Fernet encryption through a custom model field required careful handling of the encryption and decryption lifecycle across model save, retrieve, and queryset operations — including the important nuance that encrypted fields could not be filtered or sorted at the database

level, requiring application-level filtering logic for any query that needed to operate on encrypted field values. This complexity was not apparent from the encryption library's documentation and required significant debugging effort before a correct and reliable implementation was achieved.

A third technical lesson concerned the value of investing in test infrastructure early in the development cycle. The decision to write unit tests for the inference engine and classification functions in parallel with their implementation — rather than deferring testing to a dedicated testing sprint — substantially reduced the debugging effort required during integration testing, because regressions introduced by subsequent changes to the codebase were detected and isolated immediately rather than accumulating into complex compound errors.

7.3.2 Project Management Lessons

The Agile sprint plan, while providing an effective framework for managing the project's complexity, surfaced several project management lessons. The most consequential was the tendency to underestimate the time required for cross-tier integration work — specifically, the effort required to ensure that the data structures produced by the backend API were correctly consumed by the frontend components, including the handling of nested JSON structures, date-time formatting differences between Python and JavaScript, and the mapping between backend field names (snake_case) and frontend interface properties (camelCase). Retrospective analysis suggested that integration tasks should have been allocated explicit sprint capacity rather than being subsumed within the frontend and backend development story points.

A second project management lesson concerned the value of maintaining a persistent technical debt log alongside the sprint backlog. Several minor refactoring tasks — including the pagination defect identified during integration testing and the symmetry defect in the drug interaction checker — accumulated because they were identified during development but not immediately addressed. A structured technical debt log would have made these items visible and ensured their prioritisation before they were encountered as defects during the formal testing phase.

7.3.3 Domain Knowledge Lessons

The clinical domain knowledge requirements of the project provided several important lessons about the intersection of computer science and health informatics. The most important was the recognition that clinical decision logic could not be derived from general programming intuition — it required explicit engagement with authoritative clinical guidelines, a willingness to defer to clinical expertise where technical intuition conflicted with established practice, and a systematic review process in which clinical colleagues reviewed and validated the inference engine's outputs for face validity before formal testing was conducted.

The engagement with the Kenya Data Protection Act, 2019, during the security architecture design phase also provided an important lesson about the regulatory context of health technology development in Kenya. The DPA's requirements for data minimisation, purpose limitation, security safeguards, and audit logging were not merely bureaucratic constraints but reflected genuine ethical obligations to patients whose most sensitive personal data was being processed by the system. This understanding produced a more thoughtful and comprehensive security architecture than would have resulted from a purely technical approach to security design.

7.4 Conclusion

This project set out to address a central research question: could a locally grounded, AI-driven clinical decision support platform, designed in explicit alignment with Kenya's clinical guidelines, disease epidemiology, and healthcare infrastructure, meaningfully improve access to preliminary clinical guidance, chronic disease self-management, and preventive health behaviours for Kenyan patients?

The evidence produced through the design, implementation, and evaluation of SHES provided a substantive affirmative response to this question. The platform demonstrated that a clinically grounded, technically robust, and practically deployable AI health system could be built with explicit alignment to Kenya's National Clinical Guidelines, Kenya Essential Medicines List, and Data Protection Act — closing the contextualisation, comprehensiveness, transparency, and compliance gaps identified in the literature review. The forward-chaining inference engine correctly processed all fourteen clinical condition profiles and twelve emergency red flag patterns, generating clinically appropriate urgency classifications and recommendations that were independently validated as face-valid by clinical informants with relevant professional experience.

The chronic disease tracking modules — providing WHO-classified blood glucose monitoring and JNC-8-classified blood pressure monitoring with longitudinal trend visualisation — addressed one of the most consequential gaps in Kenya's primary care digital ecosystem: the absence of accessible, auto-interpreting vital sign monitoring tools for the approximately 24% of Kenyan adults living with hypertension and 2.4% living with diagnosed diabetes. The drug interaction checker, grounded in the Kenya Essential Medicines List, addressed the medication safety gap for patients on polypharmacy regimens. The mental health module addressed the documented 85%+ treatment gap in Kenya's mental health system, providing accessible, evidence-based mood tracking and coping support. The laboratory results interpreter removed a significant informational barrier between patients and the clinical significance of their own health data.

The comprehensive testing programme — encompassing 165+ test cases across unit, integration, functional, and user acceptance testing phases — produced strong evidence that the implemented system was functionally correct, technically secure, and usable by representative members of its target population. The average API response time of 187 milliseconds and the Lighthouse performance score of 87 on desktop confirmed

that the system met its performance non-functional requirements. User acceptance testing feedback from clinical and non-clinical participants provided qualitative evidence of the system's clinical utility and usability.

SHES did not claim to replace the clinical judgment of trained healthcare professionals, nor did it claim to eliminate the need for formal healthcare consultation. Rather, it was designed and demonstrated to function as a clinically intelligent first-contact tool — one that could reduce the burden of unnecessary emergency department visits for self-limiting conditions, facilitate timely care-seeking for genuinely urgent presentations, support the self-management of chronic conditions between clinical encounters, and extend the reach of evidence-based health guidance to populations and settings currently underserved by Kenya's formal healthcare system.

In conclusion, SHES represented an original, technically accomplished, and clinically grounded contribution to Kenya's health informatics landscape. It demonstrated that the intersection of artificial intelligence, modern web development, and contextualised clinical knowledge could produce a practical and impactful response to one of Kenya's most persistent public health challenges. The platform's open-source architecture, modular design, and comprehensive documentation positioned it for advancement into clinical deployment and further development by the health informatics research community — a trajectory that the researcher was committed to pursuing beyond the scope of the present project.

7.5 Recommendations

Based on the achievements, lessons learnt, and limitations identified through the SHES project, the following recommendations were made for future development. Table 7.2 presented these recommendations in order of priority, with associated technical approaches.

Table 7.2: Recommendations for Future Development of SHES

Priority	Recommendation	Technical Approach
High	Deploy SHES to a production cloud environment (AWS or Azure) with SSL certification, CDN for static assets, and automated backup	Docker Compose on cloud VM; Let's Encrypt TLS; CloudFront/Azure CDN; pg_dump scheduled backups
High	Expand the knowledge base to cover 30+ conditions, including sickle cell disease, tuberculosis, HIV-related opportunistic infections, and obstetric red flags	Extend conditions.json with additional condition objects; clinical review by specialist panel; regression test suite expansion

High	Develop a dedicated Android and iOS mobile application using React Native, sharing the same Django REST API backend	React Native project scaffolded from existing TypeScript types; offline-first capability using SQLite local cache with API sync on reconnection
High	Integrate with the Kenya Health Information System (KHIS) and KEMR to enable bidirectional patient record exchange	HL7 FHIR R4 API adapter layer; OAuth 2.0 integration with DHIS2; data mapping between SHES schema and FHIR Patient/Observation resources
Medium	Replace the rule-based inference engine with a hybrid architecture combining forward-chaining rules for emergency detection with a Naïve Bayes probabilistic classifier for condition matching, trained on anonymised Kenyan primary care morbidity data	Python Scikit-learn NaiveBayes model; training data from Kenya HMIS morbidity statistics; ensemble approach preserving rule-based emergency logic
Medium	Implement real-time telemedicine video consultation functionality connecting patients with registered clinical officers for remote consultation	WebRTC peer-to-peer video; Django Channels WebSocket signalling server; in-app consultation booking and notification system
Medium	Add a Swahili language interface to improve accessibility for users with limited English proficiency	i18n framework (react-i18next); Swahili translation file; KiSwahili clinical terminology review by language specialists
Medium	Implement push notification support for medication reminders and vital sign logging prompts	Firebase Cloud Messaging (FCM) for Android/iOS; Django Celery beat scheduler triggering notification tasks at user-configured times
Low	Add paediatric-specific clinical logic with age-stratified symptom profiles, reference ranges, and dosing guidance	Conditional knowledge base loading based on patient age; paediatric WHO growth chart integration; child-specific LAB reference ranges
Low	Conduct a formal clinical validation study comparing SHES triage outputs against clinician assessments on a set of 200+ standardised clinical vignettes representative of Kenyan primary care presentations	Prospective validation design; partnership with medical school clinical faculty; publication in a peer-reviewed health informatics journal

Source: Researcher's own analysis (2026)

Beyond the technical development recommendations detailed in Table 7.2, two broader strategic recommendations were made. First, it was recommended that the Ministry of Health Kenya consider SHES's architecture and clinical knowledge base as a reference model for the AI-powered clinical decision support component of the Kenya National eHealth Strategy, and explore opportunities to collaborate with the project developer and other academic health informatics teams in advancing the platform toward a formally evaluated public health deployment. Second, it was recommended that The Co-operative University of Kenya's School of Computing and Mathematics consider incorporating SHES as a case study in undergraduate health informatics curriculum design, demonstrating to future students how computer science skills could be applied to address Kenya's most consequential public health challenges.

7.6 Chapter Summary

This chapter presented a comprehensive conclusion to the SHES project. It evaluated the project's achievement against its five specific objectives, demonstrating full achievement of all five within the twenty-two-week development schedule. It documented the technical, project management, and domain knowledge lessons learnt through the project's execution. It provided a substantive conclusion that directly addressed the central research question, affirming based on design, implementation, and testing evidence that SHES represented a clinically meaningful, technically robust, and practically deployable contribution to Kenya's health informatics landscape. It presented a structured table of ten development recommendations across three priority levels, encompassing cloud deployment, knowledge base expansion, mobile application development, KHIS integration, probabilistic model hybridisation, telemedicine, multilingual support, push notifications, paediatric logic, and clinical validation. The chapter concluded with two strategic recommendations directed at the Kenya Ministry of Health and The Co-operative University of Kenya respectively.

END OF CHAPTER SEVEN

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APPENDICES

APPENDIX A: USER MANUAL

A.1 Introduction to the SHES User Manual

This user manual provided step-by-step guidance for all categories of SHES users — patients, doctors, and administrators — on the installation, configuration, and use of the Smart Health Expert System. The manual was written at a reading level accessible to users with secondary school education and basic digital literacy. All procedures were documented as of the system version submitted with this project report.

A.2 System Access Requirements

To use SHES, users required the following:

- A device with a modern web browser (Google Chrome version 90 or later; Mozilla Firefox version 88 or later; Microsoft Edge version 90 or later; Safari version 14 or later).
- A stable internet connection. The application was designed to function adequately on connections with a minimum download speed of 1 Mbps.
- A valid email address for account registration.
- For optimal experience, a screen width of at least 375 pixels (standard smartphone screen width) was recommended. The application was fully responsive and functional on all screen sizes from 375px upward.

No software installation was required. SHES was a browser-based web application accessible through its URL.

A.3 Account Registration

Step	Action	Expected System Response
1	Navigate to the SHES application URL in your web browser.	The Login page is displayed with a 'Register' link.
2	Click the 'Register' or 'Create Account' link on the Login page.	The Registration form is displayed.
3	Enter your full name in the 'Full Name' field.	Text is accepted in the field.
4	Enter your email address in the 'Email' field. Use a valid, accessible email address as this will be your login identifier.	Text is accepted in the field.
5	Enter a strong password of at least 8 characters in the 'Password' field. Use a mix of letters, numbers, and symbols.	Text is accepted (displayed as dots for security).
6	Re-enter your password in the 'Confirm Password' field.	Text is accepted and must match Step 5 entry.
7	Select your role from the 'Role' dropdown: 'Patient', 'Doctor', or 'Administrator'.	Role is selected and confirmed.
8	Click the 'Register' button.	If all fields are valid, your account is created and you are redirected to the Login page with a success message.
9	If an error message is displayed, read the message and correct the relevant field(s). Common errors include: email already registered; passwords do not match; password too short.	Error message identifies the specific field requiring correction.

A.4 Logging In

Step	Action	Expected System Response
1	Navigate to the SHES application URL.	The Login page is displayed.
2	Enter your registered email address in the 'Email' field.	Text is accepted.
3	Enter your password in the 'Password' field.	Text is accepted (displayed as dots).
4	Click the 'Login' button.	If credentials are correct, you are redirected to the Dashboard page. Your name and role appear in the sidebar.
5	If login fails, an error message is displayed. Check that your email and password are correct. If you have forgotten your password, contact the system administrator.	Error message is displayed for incorrect credentials.

A.5 Performing a Symptom Triage Assessment

The Symptom Triage module was the core clinical feature of SHES. It assessed your reported symptoms and provided an urgency-level recommendation to guide your health-seeking behaviour.

Step	Action	Expected System Response
1	After logging in, click 'Symptom Triage' in the left sidebar navigation menu.	The Symptom Triage page is displayed with a symptom entry form.
2	In the 'Symptom Name' field of the first symptom row, type the name of your symptom (e.g., 'fever', 'headache', 'cough').	Text is accepted.
3	Adjust the 'Severity' slider to indicate how severe the symptom feels on a scale of 1 (very mild) to 10 (extremely severe).	The numeric severity value updates in real-time as the slider is moved.
4	Enter the number of days you have had this symptom in the 'Duration (Days)' field.	Number is accepted.
5	Optionally, select a body location from the 'Body Location' dropdown.	Location is selected.
6	To add more symptoms, click the '+ Add Symptom' button and repeat Steps 2–5 for each additional symptom.	A new symptom row is added to the form for each click.
7	When all symptoms have been entered, click the 'Assess My Symptoms' button.	A loading indicator is displayed while the assessment is being processed (typically under 2 seconds).
8	Review the triage result displayed below the form. Note the urgency level badge (colour-coded: red = Emergency, amber = Doctor Visit, green = Self-Care, grey = Undetermined), the clinical recommendation text, and the matched conditions list.	Triage result is displayed with urgency badge, recommendation, matched conditions, and home-care tips.
9	If the result shows 'EMERGENCY' in red, call 999 or 0800 723 253 (Kenya Red Cross) immediately. Do not delay seeking emergency care.	Emergency contact numbers are displayed prominently in the output.
10	Your assessment is automatically saved to your Triage History. Click 'View Full History' to review all past assessments.	Triage session is saved and accessible from the Triage History page.

IMPORTANT NOTICE: The SHES symptom triage assessment was a preliminary clinical guidance tool. It did not constitute a medical diagnosis. Always seek

professional clinical advice from a qualified healthcare provider for a definitive diagnosis and treatment plan.

A.6 Logging Blood Glucose Readings

Step	Action	Expected System Response
1	Click 'Chronic Tracking' in the left sidebar.	The Chronic Disease Tracking page is displayed with tabs for Blood Glucose and Blood Pressure.
2	Ensure the 'Blood Glucose' tab is selected.	The blood glucose entry form is displayed.
3	Enter your blood glucose reading value in the 'Reading Value' field. Enter the value in mmol/L as displayed on your glucometer.	Numeric value is accepted.
4	Select the measurement context from the dropdown: 'Fasting' (before eating), 'Post-Meal' (2 hours after eating), 'Random', or 'Bedtime'.	Context is selected.
5	Click 'Save Reading'.	Reading is saved; classification label (e.g., 'Normal Fasting', 'Diabetes Range') is displayed; trend chart updates.
6	Review your classification label and trend chart. A green label indicates a normal reading. An amber or red label indicates a reading outside the target range — consult your doctor.	Classification and trend chart are displayed.

A.7 Checking Drug Interactions

Step	Action	Expected System Response
1	Click 'Medications' in the left sidebar.	The Medications page is displayed.
2	In the 'Search Medications' field, type the name of a medication you are taking. Select it from the search results dropdown.	Medication is found and can be selected.
3	Enter the dose and frequency in the provided fields. Click 'Add to My Medications'.	Medication is added to your active medication list below.
4	Repeat Steps 2–3 for all your current medications.	All medications appear in your medication list.
5	Click the 'Check Interactions' button.	The system analyses all medication pairs. An interaction report is displayed.

6	Review the interaction report. Interactions are colour-coded by severity: green = mild, amber = moderate, red = severe. Any red (severe) interactions should be discussed with your doctor or pharmacist immediately.	Interaction report with severity badges is displayed.
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A.8 Logging Out

To log out of SHES, click the 'Logout' button at the bottom of the left sidebar navigation menu. You will be redirected to the Login page, and your session token will be invalidated. For security, always log out when using a shared or public device.

APPENDIX B: DATA COLLECTION QUESTIONNAIRE

SHES Research Questionnaire: Healthcare Access and Digital Health Technology in Kenya

This questionnaire was administered as part of an undergraduate research project at The Co-operative University of Kenya. All responses were anonymous and confidential. The data collected was used solely for academic research purposes in accordance with the Kenya Data Protection Act, 2019. Completion of the questionnaire constituted informed consent to participate in the research.

Estimated completion time: 8–12 minutes.

SECTION A: DEMOGRAPHIC INFORMATION

No.	Question	Response Type
A1	What is your age group?	Single select: 18–24 / 25–34 / 35–44 / 45–54 / 55+
A2	What is your gender?	Single select: Male / Female / Prefer not to say
A3	What is your highest level of education completed?	Single select: Primary / Secondary / Diploma / Undergraduate / Postgraduate

A4	Which county do you currently reside in?	Open text
A5	What is your primary occupation?	Single select: Student / Healthcare Professional / Business / Government / Unemployed / Other

SECTION B: HEALTHCARE ACCESS AND CHALLENGES

No.	Question	Response Type
B1	How far do you live from the nearest public health facility?	Single select: <1 km / 1–5 km / 5–10 km / >10 km
B2	On average, how long do you wait to see a doctor or clinical officer at a public facility?	Single select: <30 min / 30–60 min / 1–2 hours / 2–4 hours / >4 hours
B3	I am often uncertain whether my symptoms are serious enough to visit a health facility. (1=Strongly Disagree to 5=Strongly Agree)	Likert scale 1–5
B4	I have delayed seeking medical care because I was unsure of the seriousness of my symptoms.	Single select: Never / Rarely / Sometimes / Often / Always
B5	In the past 6 months, have you managed a health complaint at home without seeing a doctor?	Single select: Yes / No
B6	If yes to B5, what was the main reason? (Select all that apply)	Multi-select: Cost / Distance / Uncertainty / Waiting times / Condition seemed mild / Other
B7	I have missed or forgotten to take prescribed medication in the past month.	Likert scale 1–5 (1=Never, 5=Very Often)
B8	I find it difficult to monitor my blood pressure or blood glucose readings regularly.	Likert scale 1–5
B9	I would benefit from a tool that explained my laboratory results in plain language.	Likert scale 1–5
B10	I have experienced challenges accessing mental health support services in Kenya.	Likert scale 1–5

SECTION C: DIGITAL TECHNOLOGY OWNERSHIP AND HEALTH APP USAGE

No.	Question	Response Type
C1	What type of device do you primarily use to access the internet?	Single select: Smartphone / Tablet / Laptop / Desktop / Feature phone / No device
C2	How would you rate your comfort level with using smartphone or web applications?	Single select: Not comfortable / Somewhat comfortable / Comfortable / Very comfortable
C3	Have you ever searched the internet for health information or medical advice?	Single select: Yes / No
C4	If yes to C3, how helpful did you find the information available online for your specific health situation in Kenya?	Single select: Not helpful / Somewhat helpful / Helpful / Very helpful / N/A
C5	Have you ever used a symptom checker or health application (e.g., WebMD, Google Health)?	Single select: Yes / No
C6	If yes to C5, please name the application(s) you have used.	Open text
C7	The health information I find online is relevant to Kenya's healthcare context and medicines.	Likert scale 1–5 (1=Strongly Disagree, 5=Strongly Agree)

SECTION D: WILLINGNESS TO USE AND FEATURE PREFERENCES

No.	Question	Response Type
D1	I would use an AI-powered application that assessed my symptoms and told me whether to see a doctor urgently.	Likert scale 1–5 (1=Extremely Unlikely, 5=Extremely Likely)
D2	I would trust an AI health application more if I knew its clinical logic was based on Kenya's official clinical guidelines.	Likert scale 1–5
D3	I would use an application that automatically classified my blood pressure or blood glucose readings against health targets.	Likert scale 1–5

D4	I would benefit from a digital medication reminder and drug interaction checker that used medications available in Kenya.	Likert scale 1–5
D5	I would use a mood tracking and mental health support feature if it was available within a health application.	Likert scale 1–5
D6	I would find it helpful to receive plain-language explanations of my laboratory test results through an application.	Likert scale 1–5
D7	I am concerned about the privacy and security of my personal health data on digital platforms.	Likert scale 1–5
D8	Which of the following features would you consider most important in a digital health application? (Rank 1–5, where 1 is most important)	Rank: Symptom assessment / Medication reminders / Vital sign tracking / Lab result explanation / Mental health support
D9	What is the maximum amount you would pay per month for access to a comprehensive digital health platform?	Single select: Nothing (free only) / KES 100–200 / KES 201–500 / KES 501–1000 / Over KES 1000
D10	Please share any additional comments, suggestions, or concerns regarding digital health services in Kenya.	Open text (optional)

Thank you for completing this questionnaire. Your responses contributed to the development of a health technology platform designed to serve Kenyan communities. The researcher greatly appreciated your time and participation.

APPENDIX C: PROPOSED PROJECT BUDGET

Table C.1 presented the itemised budget for the SHES project, covering hardware, software infrastructure, research activities, and documentation costs. All costs were expressed in Kenya Shillings (KES) and represented either actual expenditures incurred during the project or market-rate estimates for items acquired or utilised.

Table C.1: SHES Proposed Project Budget

Item	Unit Cost (KES)	Quantity	Total (KES)
Development laptop (Intel Core i5, 16GB RAM, 512GB SSD)	80,000	1	80,000
Internet subscription – fibre broadband (12 months)	2,500	12	30,000
Cloud hosting – AWS EC2 t3.small instance (12 months)	2,800	12	33,600
Domain name registration (.co.ke, 2 years)	2,500	1	2,500
SSL certificate (Let's Encrypt – free tier)	0	1	0
PostgreSQL cloud backup storage (AWS S3, 12 months)	500	12	6,000

Clinical guidelines documents and reference materials	3,500	1	3,500
Research consumables (printing, stationery, binding)	4,500	1	4,500
Data collection – questionnaire printing and distribution	2,000	1	2,000
Transportation for fieldwork observation sessions	1,500	2	3,000
Participant refreshments during UAT sessions	500	8	4,000
Report printing and spiral binding (draft copies)	350	5	1,750
Final report hardcover binding and printing	1,500	3	4,500
Contingency reserve (10%)	—	—	17,535
TOTAL PROJECTED BUDGET			KES 192,885

Source: Researcher's own compilation based on market rates and actual expenditures (2026)

Notes: Cloud hosting costs were estimated based on AWS EC2 t3.small instance pricing in the af-south-1 (Cape Town) region, which provided the lowest latency for Kenyan users. Internet costs were based on the Safaricom Home Fibre 20 Mbps plan rate as of January 2024. The contingency reserve of 10% was applied to cover unforeseen costs including additional cloud compute usage during testing and peer review printing costs. Open-source software tools — including Django, React, PostgreSQL, Docker, and all supporting libraries — were used at no licensing cost under their respective open-source licences.

APPENDIX D: PROJECT SCHEDULE

Table D.1 presented the SHES project schedule in Gantt chart format, illustrating the distribution of development activities across the twenty-two-week project timeline. Shaded cells (shown in blue) indicated the weeks during which each sprint's activities were actively in progress.

Table D.1: SHES Project Gantt Chart Schedule

Sprint	Activity	Wk 1-2	Wk 3-4	Wk 5-10	Wk 11-15	Wk 16-18	Wk 19-22
1	Requirements gathering, stakeholder interviews, questionnaire						
2	System design: architecture, ERD, wireframes, SRS documentation						
3	Backend setup: Django project, DB schema, JWT authentication						
4	Inference engine: knowledge base design, forward-chaining logic, unit tests						
5	Medications module, drug interaction checker, chronic tracking modules						

6	Mental health module, lab results interpreter, patient profile module						
7	React frontend: all 9 pages, TanStack Query integration, Recharts						
8	Integration testing, UAT, security hardening, defect resolution						
9	Docker deployment config, final documentation, report writing						

Source: Researcher's own project schedule design (2026)

The project schedule followed the nine-sprint Agile development plan documented in Chapter Three. Sprint 4 (Weeks 5–10) was the longest sprint, allocated six weeks to accommodate the clinical knowledge base design and inference engine implementation — the most intellectually demanding and time-consuming components of the project. The parallel overlap of backend module development (Sprints 4–6) with architecture and database design (Sprint 2) reflected the iterative nature of Agile development, where detailed module design emerged progressively as high-level architectural decisions were finalised. Sprint 8 (Weeks 19–20) was dedicated exclusively to integration and user acceptance testing to ensure that the testing phase received undivided attention without competing development commitments.

APPENDIX E: STAKEHOLDER INTERVIEW GUIDE

The following semi-structured interview guide was used during the requirements gathering phase of the SHES project. Interviews were conducted with clinical officers, registered nurses, and health facility administrators. The guide served as a thematic framework for the conversation; actual interviews were conducted flexibly, following the direction of the interviewee's responses while ensuring all thematic areas were covered.

Interview Preamble

"Thank you for agreeing to speak with me today. I am a final-year Software Engineering student at The Co-operative University of Kenya. I am developing an AI-powered health application called SHES — the Smart Health Expert System — which aims to provide Kenyan patients with preliminary symptom assessment, chronic disease monitoring, and preventive health guidance. Your professional insights and experience will be invaluable in shaping the design of this system. This interview should take approximately 45–60 minutes. With your permission, I would like to record our conversation for transcription purposes. All responses will be kept strictly confidential, and no personally identifying information will appear in my report. Do you consent to participate and to the recording of this interview?"

Theme 1: Current Clinical Workflow and Triage Practices

1. Can you walk me through a typical patient's journey from arrival at the outpatient department to seeing a clinician? Where does triage occur, and how is urgency determined?
2. What tools or protocols, if any, are currently used to perform patient triage at this facility? How consistent is the triage process across different staff members?
3. Approximately what proportion of outpatient presentations would you estimate are non-urgent — that is, conditions that could have been managed through self-care or a simple phone consultation?
4. How does patient volume at peak times affect the quality of clinical consultation? What happens to waiting times and consultation duration?

Theme 2: Chronic Disease Management Challenges

5. In your experience, how well do patients with hypertension or diabetes manage their conditions between facility visits? Do you observe consistent self-monitoring of blood pressure or blood glucose?
6. How frequently do patients present with acute complications of chronic conditions — such as hypertensive crises or hypoglycaemic episodes — that you believe were preventable with better self-monitoring?
7. What are the main barriers you observe to effective medication adherence among your chronic disease patients?

Theme 3: Digital Health Technology Perspectives

8. Are you aware of any digital health tools currently used by patients or recommended by this facility? If so, what are their strengths and limitations in your view?
9. If a patient-facing AI health application were available that provided symptom assessment, chronic disease tracking, and medication reminders, what features would you consider most clinically important?
10. What concerns, if any, would you have about patients using an AI-powered symptom checker? How would you suggest those concerns be addressed in the system's design?
11. What data privacy and security standards would you consider essential for a health application handling patient health data in Kenya?

Theme 4: Closing Questions

12. Is there anything else about Kenya's primary care challenges or digital health needs that you believe is important and that we have not covered?
13. Would you be willing to be contacted for a follow-up conversation during the system testing phase to review the application's clinical outputs?

"Thank you very much for your time and insights. Your contribution will make a real difference to the design of this system."

APPENDIX F: INFERENCE ENGINE KNOWLEDGE BASE SAMPLE

The following JSON excerpt illustrated the structure of the SHES clinical knowledge base as implemented in the conditions.json file. The excerpt showed three of the fourteen condition objects: Malaria, Common Cold, and Hypertension.

```
{
  "conditions": [
    {
      "id": "malaria",
      "name": "Malaria",
      "required_symptoms": ["fever", "chills"],
      "supporting_symptoms": ["headache", "sweating",
"fatigue",
                                "body aches", "nausea",
"vomiting"],
      "match_threshold": 0.5,
      "default_urgency": "DOCTOR_VISIT",
      "clinical_source": "Kenya National Clinical Guidelines,
2016"
    },
    {
      "id": "common_cold",
      "name": "Common Cold",
      "required_symptoms": ["runny nose"],
      "supporting_symptoms": ["sore throat", "cough",
"sneezing",
                                "mild fever", "congestion",
"fatigue"],
      "match_threshold": 0.4,
      "default_urgency": "SELF_CARE",
      "clinical_source": "WHO Primary Care Protocols, 2020"
    },
    {
      "id": "hypertension",
      "name": "Hypertension",
      "required_symptoms": ["headache", "high blood pressure"],
      "supporting_symptoms": ["dizziness", "blurred vision",
"chest
                                tightness", "nosebleed",
"fatigue"],
      "match_threshold": 0.5,
      "default_urgency": "DOCTOR_VISIT",
      "clinical_source": "Kenya National Clinical Guidelines,
2016;
                                JNC-8 Guidelines, 2014"
    }
  ]
}
```

The red_flags.json file contained the twelve emergency pattern definitions. The following excerpt illustrated two of the twelve patterns:

```

{
  "red_flags": [
    {
      "id": "chest_pain_cardiac",
      "name": "Suspected Cardiac Event",
      "keywords": ["chest pain", "chest tightness", "radiating
arm pain",
                  "crushing chest", "heart attack"],
      "emergency_guidance": "Call 999 or Kenya Red Cross 0800
723 253
                           immediately. Do not drive
yourself."
    },
    {
      "id": "stroke_signs",
      "name": "Suspected Stroke (FAST Signs)",
      "keywords": ["face drooping", "arm weakness", "speech
difficulty",
                  "sudden confusion", "sudden vision loss",
"stroke"],
      "emergency_guidance": "Call 999 immediately. Note the
time symptoms
                           started. Do not give food or
water."
    }
  ]
}

```

END OF APPENDICES